

1 The multiscale architecture of personal 2 light exposure

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34 Abstract

35 Personal light exposure has been associated with sleep and non-communicable diseases,
36 yet how ocular exposure varies across the day, among people and between everyday
37 settings remains poorly understood. We analysed 191 adults at nine sites in seven countries;
38 184 contributed 1,478 participant-days (near eye: 141 adults, 816 days; complementary
39 chest: 154 adults, 902 days). The shared 24-hour near-eye pattern accounted for 78.9% of
40 full-model R^2 , participant patterns 12.9%, participant-day shifts 6.2% and site patterns 2.0%.
41 Reported light source and immediate setting produced large contrasts, accounting for 24.2%
42 and 30.7%, respectively, of variation in fitted hourly patterns, the next-largest shares after
43 time of day. Site, civil-photoperiod and person-level associations were metric-specific rather
44 than uniform. Only 24.0% of daytime minutes met the recommendation. By resolving
45 variation across sites, people, days and immediate settings, this study lays an empirical
46 foundation for targeted, harmonised exposure monitoring aligned with health-based light
47 recommendations.

48 Light reaching the eye supports vision and signals circadian timing, sleep, alertness and
49 neuroendocrine responses¹⁻⁵. Melanopic equivalent daylight illuminance (melanopic EDI)
50 weights illuminance for melanopsin-related sensitivity and is most relevant to these effects^{6,7}.
51 Expert consensus recommends at least 250 lx melanopic EDI at the eye during daytime, no
52 more than 10 lx during the three hours before sleep and no more than 1 lx in the sleep
53 environment for healthy adults with regular daytime schedules⁸. These physiology-based
54 ranges make daily light a health-relevant exposure, but are not individual risk functions.

55 Population studies and reviews associate personal light patterns with sleep and major non-
56 communicable diseases⁹. Wrist-sensor studies link light timing and day-night distribution with
57 sleep timing and sleepiness, psychiatric disorders, incident type 2 diabetes, cardiovascular
58 disease and mortality¹⁰⁻¹⁵. Other cohorts link evening, nocturnal or seasonal light with sleep
59 in bipolar disorder and with obesity, diabetes, hypertension and metabolic health¹⁶⁻¹⁹. The
60 exposome framework characterises non-genetic exposures across the life course^{20,21}, yet
61 rarely includes time-resolved ocular melanopic light. Existing studies establish population
62 relevance but mostly measured broad-spectrum light at the wrist or bedside, providing only
63 proxy evidence for melanopic ocular exposure.

64 An everyday exposure baseline against these recommendations is missing. Personal
65 exposure to natural and electric light follows an expected 24-hour rhythm, including in large
66 wrist-sensor cohorts²², but previous studies focus on particular populations, occupations or

67 settings²³⁻²⁵. Its amplitude and timing across sites and countries, and the contributions of
68 time, site, individuals, day-to-day variation and immediate context, remain uncertain.
69 Daylight availability, satellite estimates and fixed-site measurements cannot resolve these
70 layers as people move through buildings, transport, outdoor spaces and sleep
71 environments²⁶. Sensor position also matters: near-eye sensing is closer to the incident
72 ocular field but is not a retinal measurement; wrist and chest sensors capture related fields
73 shaped by posture, wear and occlusion²⁷⁻³³.

74 Using the harmonised MeLiDos protocol, we measured personal light across nine sites in
75 seven countries^{34,35}. The near-eye sensor characterised ocular light exposure during wear,
76 alongside diaries, light-source and immediate-setting reports, sleep information and
77 questionnaires. The complementary chest sensor provided non-ocular environmental
78 evidence and enabled participation among people reluctant to wear the glasses-mounted
79 sensor. We sought to establish a multisite baseline against the recommendation windows
80 and quantify contributions from local-clock pattern, site and photoperiod, participant and
81 participant-day, immediate environment and routine, and selected person-level correlates.
82 These nested contributions define the multiscale architecture of personal light exposure.
83 Although not global, the protocol enables harmonised measurement elsewhere,
84 progressively closing gaps across locations, civil photoperiods, latitudes and climates and
85 building towards a global account of ocular light exposure.

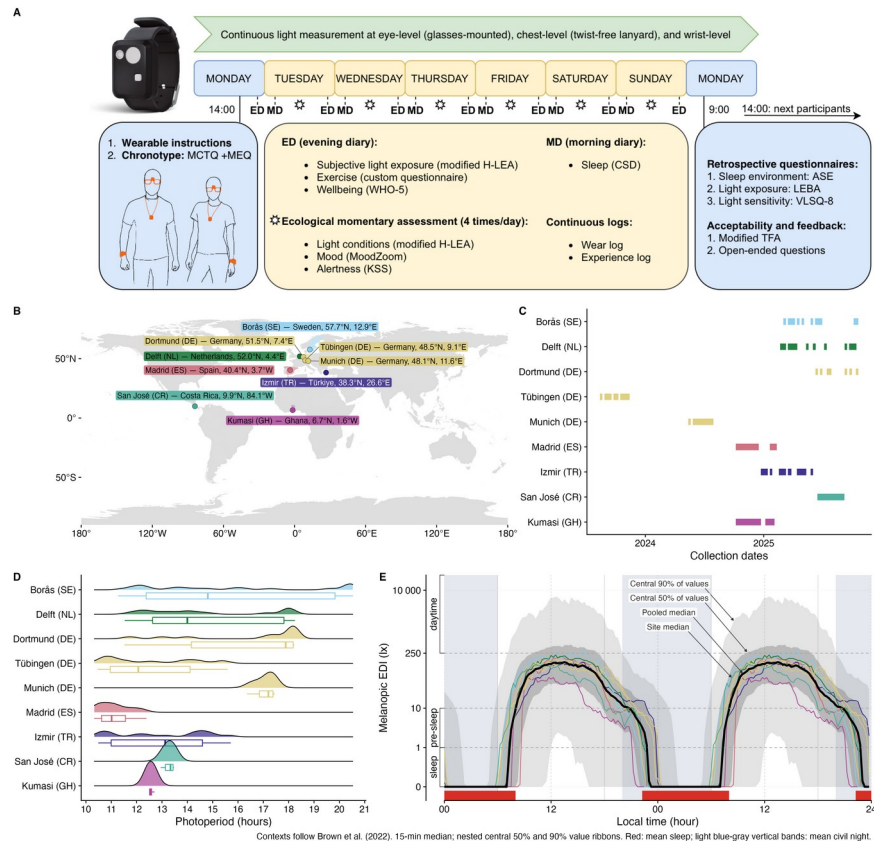
86 Results

87 A multisite baseline and recommendation adherence

88 The normalised roster comprised 191 participants across Borås (SE), Delft (NL), Dortmund
89 (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San José (CR) and Kumasi
90 (GH). Of these, 184 contributed 1,478 participant-days with recorded, non-all-zero light data
91 at either position before quality screening. After screening, the near-eye dataset contained
92 141 participants and 816 participant-days, the complementary chest dataset 154 and 902,
93 and the common sample 112 participants and 643 participant-days. Near-eye
94 measurements provide ocular-exposure evidence during wear, whereas chest
95 measurements are non-ocular; both describe the bedside sleep environment during reported
96 sleep. [Supplementary Table S1](#) gives the sample flow.

97 The pooled near-eye record followed the expected 24-hour rhythm: low overnight exposure,
98 a morning rise, higher daytime values and an evening decline. Seventeen complementary
99 metrics captured thematic streams rather than one score. Level metrics described daily

100 geometric mean melanopic EDI and the brightest and darkest supported 10-hour means;
101 duration and timing metrics quantified exposure ranges, sustained bright periods, bright or
102 dark windows and threshold crossings. Dose integrated intensity, regularity captured stability
103 and fragmentation, and melanopic daylight efficacy (MDER) described spectral composition
104 relative to visual illuminance. These features are not interchangeable health-response
105 estimates. [Supplementary Table S2](#) and [Supplementary Figure S1](#) give the metric dictionary
106 and site distributions; [Supplementary Figure S2](#) shows a worked derivation. [Figure 1](#) and
107 [Table 1](#) summarise the design and data structure.



108

109 *Figure 1: Five-panel overview of the eight-day protocol, sensor positions, nine*
 110 *country-coded sites, collection windows, civil photoperiod, and the pooled near-eye*
 111 *daily profile. A, Eight-day field protocol and sensor positions. The wrist logger is*
 112 *shown as a protocol device but is not an exposure channel analysed in this*
 113 *manuscript. B, Nine sites in seven countries. C, Site collection periods. D, Observed*
 114 *civil photoperiod. E, Site and pooled 15-minute median near-eye melanopic*
 115 *equivalent daylight illuminance profiles from 816 participant-days contributed by 141*
 116 *participants, repeated over 48 hours for display only. Grey ribbons show the central*
 117 *50% and 90% of measured values, not confidence intervals; coloured lines are site*
 118 *medians, the black line is the pooled median, and red and blue shading show mean*
 119 *reported sleep and civil night (sun elevation < -6°). During reported sleep, the*
 120 *measurement represents the bedside sleep environment.*

121

122 *Table 1: Participant and measurement characteristics overall and by country-coded study site. The table distinguishes the 191-person roster,*
123 *participants and participant-days with recorded light data, and the screened near-eye and complementary chest samples. Model-specific samples*
124 *are reported with each analysis. Of the 191 roster participants, 184 contributed the 1,478 recorded participant-days. Continuous participant*
125 *characteristics are median with the stated percentile range.*

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Site specifics										
Institution	Overall	RISE	THUAS	BAuA	MPI	TUM	FUSPCEU	IZTECH	UCR	KNUST
Country	—	Sweden	Netherlands	Germany	Germany	Germany	Spain	Türkiye	Costa Rica	Ghana
City	—	Borås	Delft	Dortmund	Tübingen	Munich	Madrid	Izmir	San José	Kumasi
Coordinates	—	57.7°N, 12.9°E	52.0°N, 4.4°E	51.5°N, 7.4°E	48.5°N, 9.1°E	48.1°N, 11.6°E	40.4°N, 3.7°W	38.3°N, 26.6°E	9.9°N, 84.1°W	6.7°N, 1.6°W
Data collection										
Participants ¹	191 roster (g:143; c:157; p:116)	17 roster (g:14; c:17; p:14)	20 roster (g:13; c:15; p:13)	24 roster (g:19; c:22; p:19)	26 roster (g:26; c:0; p:0)	10 roster (g:10; c:10; p:10)	23 roster (g:23; c:22; p:22)	17 roster (g:17; c:17; p:17)	39 roster (g:6; c:39; p:6)	15 roster (g:15; c:15; p:15)
Participant-days ¹	1478 roster (g:1134 c:1246; p:902)	137 roster (g:107 c:137; p:107)	125 roster (g:107 c:124; p:106)	176 roster (g:145 c:163; p:132)	208 roster (g:208 c:0; p:0)	80 roster (g:80 c:80; p:80)	182 roster (g:182 c:174; p:174)	138 roster (g:138 c:138; p:138)	312 roster (g:48 c:312; p:48)	120 roster (g:119 c:118; p:117)
Participant time ²	116w 4d	11w 23h	11w 23h	15w 2d	21w 3d	8w 4d	18w 3d	14w 3d	4w 4d	11w 4d
Declared non-wear ²	g:4w 1d (3.66%)	g:2d 14h (3.34%)	g:3d 1h (3.92%)	g:3d 15h (3.42%)	g:6d 10h (4.30%)	g:2d 17h (4.55%)	g:2d 20h (2.20%)	g:3d 3h (3.13%)	g:1d 11h (4.66%)	g:3d 21h (4.80%)
Screened days ^{1,3}	g:816 c:902	g:78 c:96	g:78 c:93	g:107 c:114	g:150 c:0	g:60 c:60	g:129 c:123	g:101 c:102	g:32 c:230	g:81 c:84
Civil photoperiod ^{4,2}	13.25 h (10.62–18.20) (n=1478 d)	14.81 h (11.80–20.49) (n=137 d)	14.00 h (11.88–18.09) (n=125 d)	17.88 h (11.76–18.22) (n=176 d)	12.07 h (10.55–15.41) (n=208 d)	17.20 h (16.45–17.45) (n=80 d)	11.01 h (10.35–12.22) (n=182 d)	13.13 h (10.52–15.57) (n=138 d)	13.34 h (13.01–13.47) (n=312 d)	12.54 h (12.49–12.68) (n=120 d)

Table continued

	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)
Participant information										
Age ⁴	28 y (21 y–52 y) (n=191 N)	38 y (22.6 y–64.8 y) (n=17 N)	30.5 y (19.9 y–57.1 y) (n=20 N)	35.5 y (20.1 y–58.5 y) (n=24 N)	27 y (22 y–37 y) (n=26 N)	27.5 y (22.2 y–29 y) (n=10 N)	31 y (20.2 y–54.3 y) (n=23 N)	24 y (21 y–29.6 y) (n=17 N)	34 y (22 y–48.1 y) (n=39 N)	23 y (20.1 y–25 y) (n=15 N)
Sex ⁵	F:105 / M:86 (n=191)	F:6 / M:11 (n=17)	F:8 / M:12 (n=20)	F:13 / M:11 (n=24)	F:14 / M:12 (n=26)	F:6 / M:4 (n=10)	F:15 / M:8 (n=23)	F:11 / M:6 (n=17)	F:24 / M:15 (n=39)	F:8 / M:7 (n=15)
Employment status ⁶	F:163 / P:26 / N:2 (n=191)	F:16 / P:0 / N:1 (n=17)	F:11 / P:9 / N:0 (n=20)	F:21 / P:3 / N:0 (n=24)	F:23 / P:3 / N:0 (n=26)	F:9 / P:1 / N:0 (n=10)	F:21 / P:2 / N:0 (n=23)	F:15 / P:2 / N:0 (n=17)	F:34 / P:5 / N:0 (n=39)	F:13 / P:1 / N:1 (n=15)
Chronotype group ⁷	m:62 / i:94 / e:30 (n=186)	m:8 / i:4 / e:5 (n=17)	m:3 / i:9 / e:3 (n=15)	m:8 / i:13 / e:3 (n=24)	m:9 / i:9 / e:8 (n=26)	m:0 / i:8 / e:2 (n=10)	m:5 / i:17 / e:1 (n=23)	m:2 / i:10 / e:5 (n=17)	m:16 / i:20 / e:3 (n=39)	m:11 / i:4 / e:0 (n=15)
Sleep-corrected midsleep on free days ^{4,8}	04:03 (02:08–06:15) (n=185 N)	02:55 (02:04–05:09) (n=17 N)	04:31 (03:05–05:37) (n=15 N)	03:47 (02:17–04:49) (n=24 N)	04:33 (02:41–06:04) (n=26 N)	05:08 (04:12–06:04) (n=10 N)	04:48 (03:10–06:52) (n=22 N)	05:29 (03:35–06:49) (n=17 N)	03:42 (02:17–05:31) (n=39 N)	02:19 (01:22–03:25) (n=15 N)

¹ g, near-eye glasses position; c, complementary chest position; p, paired common sample in which the same participant-day is available at both positions.

² w: weeks; d: days; h: hours; min: minutes. Participant time and declared non-wear use eligible real minutes on screened days; both are shown for near eye only.

³ Screened days are the final participant-days after the 80% completeness screen and exclusion of exact-all-zero melEDI days.

⁴ Median (5th percentile, 95th percentile).

⁵ Sex categories: Female and Male.

⁶ Employment categories: Full/studying, Part/marginal, and Not employed.

⁷ Chronotype groups follow the Morningness–Eveningness Questionnaire score.

⁸ MCTQ: Munich Chronotype Questionnaire; MEQ: Morningness–Eveningness Questionnaire. Clock summaries are circular.

128 Exposure was within the applicable recommendation during 24.0% of daytime minutes,
129 63.3% of pre-sleep minutes and 87.7% of sleep minutes. This represented 137,792 of
130 573,712 waking daytime minutes, 81,894 of 129,390 pre-sleep minutes and 336,052 of
131 383,366 minutes during reported sleep. The sleep quantity describes the bedside sleep
132 environment because the device was not worn at the eye; the Discussion addresses low-
133 light measurement accuracy. [Supplementary Table S3](#) gives exact numerators,
134 denominators and complementary ranges.

135 These pooled-minute percentages describe how often exposure lay within range, not
136 participant adherence. Model-based comparisons retained each window's adherent and
137 valid-minute counts.

138 The main adherence model included 2,216 recommendation-window periods from 140
139 participants and 782 complete daytime, pre-sleep and sleep cycles, representing 1,028,958
140 valid minutes. Site-average work-day adherence was 26.5% during daytime, 61.9% during
141 pre-sleep and 90.3% during sleep; free-day adherence was 21.6%, 69.9% and 84.1%.
142 Relative to work days, free-day adherence was 4.92 percentage points lower during daytime
143 (95% confidence interval (95% CI) 2.21 to 7.64 lower), 7.98 higher during pre-sleep (3.28 to
144 12.67) and 6.13 lower during sleep (3.48 to 8.79). Requiring at least 80% valid coverage
145 preserved each direction and interval conclusion, with a maximum 0.88-point change.

146 Adherence differences between free and work days varied among sites ($p = 0.003$): during
147 daytime ($p = 0.002$) and sleep ($p = 0.003$), but not pre-sleep ($p = 0.526$). The free-minus-
148 work daytime difference exceeded its site-average value by 15.0 percentage points in
149 Dortmund (DE; 95% CI 6.9 to 23.1; $p = 0.004$) and was 9.2 lower in Madrid (ES; -15.6 to
150 -2.8; $p = 0.042$); the sleep difference was 6.3 higher in Kumasi (GH; 3.4 to 9.1; $p < 0.001$).
151 These directions persisted when the sensitivity analysis required at least 80% valid data
152 coverage.

153 Fixed predictors explained 58.4% of variance on the design-standardized response scale,
154 the participant intercept added 2.2 percentage points and observation and distribution
155 variation accounted for 39.4%; conditional R^2 was 60.6%. Within marginal R^2 , Shapley
156 shares were 94.1% for recommendation window, 4.8% for site and 1.2% for day type. On the
157 design-standardized response scale, site accounted for 2.8 percentage points of variance,
158 compared with a 2.2-point participant-intercept increment and 0.7 percentage points for day
159 type. Thus, the site allocation was about 1.3 times the participant-intercept increment and
160 about four times the day-type allocation. [Table 2](#) summarises the main estimates;
161 [Supplementary Figures S4](#) and [S5](#) show site-average and site-specific results.

162 Table 2: Recommendation adherence by Brown et al. recommendation window and day type. Daytime, Pre-sleep and Sleep identify the
163 recommendation windows used for comparison. Daytime comprises diary-defined wake excluding the three hours before Sleep, Pre-sleep
164 comprises those three hours, and Sleep uses the reported sleep interval and therefore describes the bedside sleep environment. Pooled minutes
165 meeting recommendation report the number and percentage of valid measured minutes whose exposure lay within the applicable range. Work-
166 day and free-day percentages, differences, 95% confidence intervals and false-discovery-rate-adjusted p-values are site-average estimates from
167 the main recommendation-window model, with each of the nine sites weighted equally. The pooled-minute fractions and modelled
168 recommendation-window estimates are distinct estimands. The main model included 2,216 recommendation-window periods from 140
169 participants and 782 behavioural cycles, representing 1,028,958 valid minutes. Adherence is defined at the measured-minute or recommendation-
170 window-period level and does not classify participants.

Recommendation adherence by recommendation window and day type

	Recommendation	Observed minute-level adherence	Site-average recommendation-window model			
		Minutes meeting recommendation, n/N (%)	Work day, % (95% CI)	Free day, % (95% CI)	Free minus Work, percentage points (95% CI)	FDR-adjusted p
Daytime	At least 250 lx melanopic EDI during wake, excluding the three hours before sleep	137,792/573,712 (24.0%)	26.5 (24.4 to 28.6)	21.6 (19.3 to 23.9)	-4.9 (-7.6 to -2.2)	<0.001
Pre-sleep	No more than 10 lx melanopic EDI during the three hours before reported sleep	81,894/129,390 (63.3%)	61.9 (58.5 to 65.4)	69.9 (66.3 to 73.5)	+8.0 (+3.3 to +12.7)	<0.001
Sleep	No more than 1 lx melanopic EDI during reported sleep; bedside sleep environment	336,052/383,366 (87.7%)	90.3 (88.7 to 91.8)	84.1 (81.6 to 86.7)	-6.1 (-8.8 to -3.5)	<0.001

Observed pooled-minute fractions and site-average model estimates are distinct estimands. Each of the nine sites contributes equally to the model estimates.

172 A separate exploratory analysis related adherence across recommendation windows. Within
173 participants, the estimated differences for a daytime-adherence value 10 percentage points
174 above a participant's usual level were -0.26 percentage points for sleep adherence (95% CI
175 -0.95 to 0.43) and -1.09 points for pre-sleep adherence (-2.75 to 0.56). Both intervals
176 included zero, and temporal dependence remained unresolved, so we withheld a within-
177 participant day-level claim rather than interpreting these estimates as evidence of no
178 association. Between participants, each 10 percentage points higher observed average
179 daytime adherence was associated with 2.52 points lower sleep adherence (95% CI 1.48 to
180 3.55 lower; FDR-adjusted $p < 0.001$) and 3.59 points lower pre-sleep adherence (1.25 to
181 5.94 lower; FDR-adjusted $p = 0.005$). Directions and interval conclusions remained in the
182 sensitivity analysis, with a maximum 0.60-point change. Thus, participants with higher
183 average daytime adherence tended to have lower pre-sleep and sleep adherence,
184 consistent with higher evening and nighttime light exposure. Complete estimates are
185 reported in [Supplementary Table S4](#), with participant profiles in [Supplementary Figure S6](#).

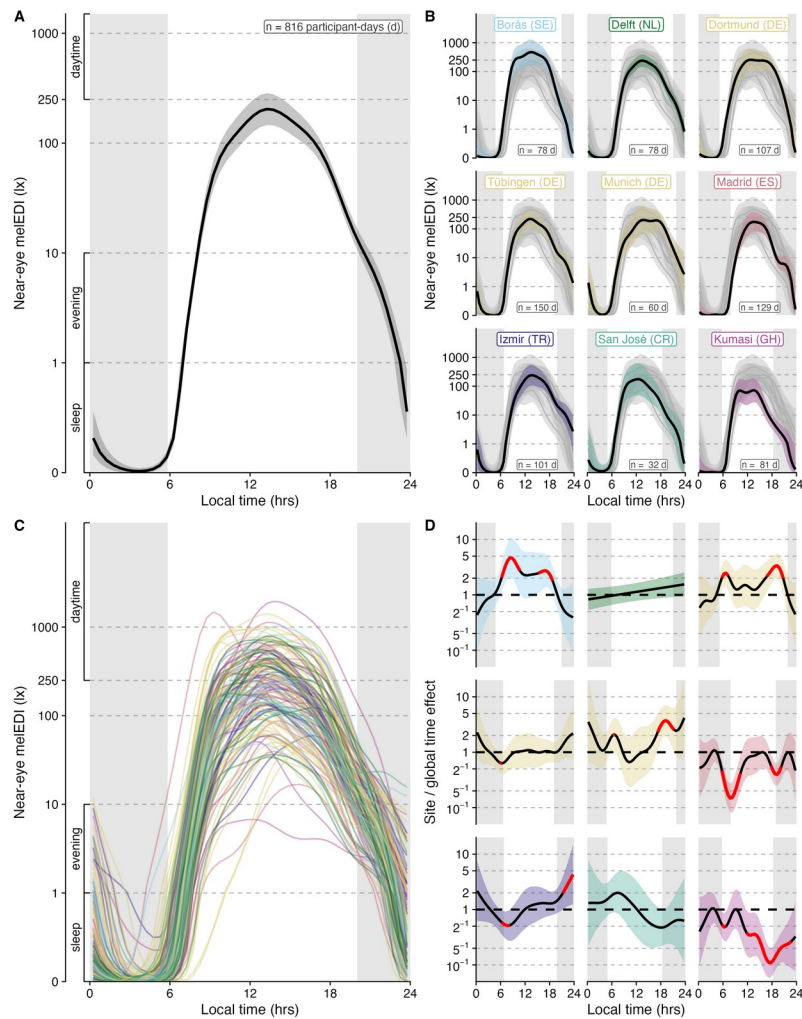
186 **Personal and day-to-day differences exceed site** 187 **differences**

188 A nonlinear model decomposed personal light exposure into common local-clock, site and
189 participant curves, and participant-day shifts. Near-eye in-sample R^2 was 0.775. Participant
190 curves were 1.80 times as dispersed as site curves (95% CI 1.16 to 4.33); participant curves
191 plus day shifts had a dispersion ratio of 1.99 (1.29 to 4.77). After accounting for time of day,
192 fitted variation among people and their days within sites exceeded variation among sites.

193 Shares of full-model R^2 showed the same hierarchy: 78.9% for the shared local-clock
194 pattern, 12.9% for participant patterns, 6.2% for participant-day shifts and 2.0% for site
195 patterns. The participant-pattern share was 6.41 times the site-pattern share (3.85 to 15.14),
196 and the combined participant-pattern and participant-day-shift share was 9.49 times the site-
197 pattern share (5.77 to 21.81).

198 The shared rhythm did not imply uniform amplitude. Relative to the fitted site-average daily
199 pattern, Borås (SE) ratios were 2.04 to 4.64 from 06:00 to 11:00 and 1.80 to 2.73 from 15:00
200 to 19:00. Madrid (ES) ratios were 0.15 to 0.47 from 05:30 to 10:30 and 0.40 to 0.57 from
201 17:30 to 20:30. Izmir (TR) shifted from 0.51 to 0.59 at 06:00 to 08:30 to 1.98 to 4.28 at 21:00
202 to 24:00, while Kumasi (GH) was 0.11 to 0.38 from 11:30 to 23:00. These are clock-specific
203 departures from the average curve.

204 The chest model had an in-sample R^2 of 0.737. Of this full-model R^2 , common time
205 accounted for 79.1%, participant patterns 11.9%, participant-day shifts 7.0% and site
206 patterns 2.1%. Common-sample comparisons supported the combined participant-plus-day
207 ordering, but the participant-only contrast was less stable. Cross-position similarity is
208 complementary context, not replication, equivalence or interchangeability. [Figure 2](#) shows
209 the fitted layers; [Supplementary Tables S5](#) and [S6](#) give both decompositions.



210

211 *Figure 2: Multiscale daily pattern of near-eye melanopic equivalent daylight*
 212 *illuminance. A, Site-average daily curve. B, Fitted site curves. C, Fitted participant*
 213 *curves after the shared local-clock and site patterns were considered. D, Ratio of*
 214 *each fitted site curve to the site-average curve on the melanopic equivalent daylight*
 215 *illuminance plus 0.1 lx scale. Grey bands show mean civil night. Shaded bands are*
 216 *clock-specific 95% confidence intervals; red segments identify displayed half-hours*
 217 *whose interval did not include the site-average curve and are descriptive clock-*
 218 *specific contrasts. The analysis included 141 participants and 816 participant-days*
 219 *across nine sites. The label “global time effect” denotes the site-average local-clock*
 220 *curve (panel A).*

221 Geography and photoperiod provide partial context

222 Site associations were stream-specific. Site differences met the FDR criterion for 8 of 17
 223 near-eye metrics: all three level metrics, waking time above 250 lx, the brightest- and

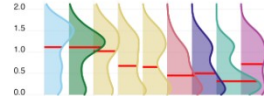
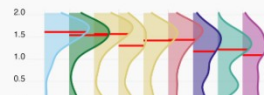
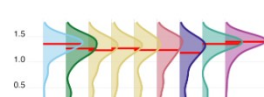
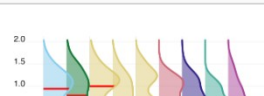
224 darkest-10-hour midpoints, and the mean and last timing above 250 lx. They were not FDR-
225 retained for dose, spectral composition, stability, fragmentation, sustained bright exposure,
226 the other threshold-duration metrics or first timing above 250 lx. Across metric models,
227 conditional R^2 averaged 36.4%; participant-associated R^2 averaged 25.6% among
228 participant-day metrics. Among the eight metrics with site support, site-associated part R^2
229 averaged 9.0%. These non-additive components reinforce the stronger individual-level
230 contribution in the daily-pattern analysis.

231 Civil-photoperiod associations were selective. Per additional hour, the fitted daily geometric
232 mean was 1.18 times higher (95% CI 1.11 to 1.26), the brightest-10-hour mean 1.24 times
233 higher (1.13 to 1.36), waking time above 250 lx 1.14 times higher (1.07 to 1.20) and daily
234 dose 1.28 times higher (1.17 to 1.39). Each hour was also associated with 0.134 h less time
235 below 10 lx before sleep (0.066 to 0.201 h less) and the last exposure above 250 lx
236 occurring 0.323 h later (0.186 to 0.460 h). Civil photoperiod met the FDR criterion for 12 of
237 17 metrics but not stability, fragmentation, sleep time below 1 lx, the brightest-window
238 midpoint or first exposure above 250 lx. Among supported models, photoperiod-associated
239 part R^2 averaged 5.2%; the corresponding mean for the six supported latitude models was
240 4.0%, meaning that photoperiod had a stronger effect than latitude, when compared across
241 models.

242 Across the recorded 10.33 to 20.52 h of civil photoperiod, six of nine level, exposure-history,
243 and duration-based metrics changed from increasing to a sustained near-flat tail between
244 14.24 and 16.20 hours of photoperiod: the daily mean, both 10-hour window means, time
245 above 1,000 lx, waking time above 250 lx and dose. Pre-sleep time below 10 lx, sleep-
246 environment time below 1 lx and the longest period above 250 lx did not. This transition
247 means that the effect of photoperiod becomes uncertain around 14 to 16 hours. The fitted
248 slope became compatible with zero and remained so through the observed maximum. Some
249 classifications were sensitive to site, sample or model form. Several limitations apply as
250 discussed below: latitude was confounded with site, collection periods and photoperiod
251 overlap differed, and weather, built environment, culture and routine could not be separated.
252 The metric-specific results are summarised in [Table 3](#), [Supplementary Figure S7](#) and
253 [Supplementary Tables S7](#) and [S8](#).

254 Table 3: Near-eye personal light-exposure metrics and their geographic and photoperiod context. Descriptive exposure values are medians with
255 interquartile ranges (25th to 75th percentiles). Density plots show site distributions and are descriptive. Metrics are grouped into thematic streams
256 and describe distinct features of the daily exposure pattern rather than interchangeable health-response estimates. Model cells report overall-site
257 evidence, civil-photoperiod associations with 95% confidence intervals, FDR decisions, participant-associated model information and model-
258 specific R^2 summaries where applicable. Site and photoperiod part- R^2 values may contain overlapping fitted information and must not be summed.
259 Associations are observational. Absolute latitude is fixed within and confounded with study site, so separate models containing either site or
260 latitude are reported; complete latitude and model-adequacy results are provided in the Supplementary Information.

Near-eye personal light-exposure metrics and their geographic and photoperiod context

		Descriptive summary		Association evidence		Modelled variation
Definition/relevance		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Duration						
Time above 1,000 lx meEDI	Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	Unit h Median 0.683 IQR 0.183 to 1.558 Participants 141 Days 816		FDR not supported; adjusted p 0.071 Part R-squared 4.7% (2.6% to 12.1%)	Estimate Ratio per 1 h: 1.206 (1.128 to 1.288) FDR supported; adjusted p <0.001 Part R-squared 9.0% (3.7% to 15.7%)	Marginal 21.5% (16.3% to 32.3%) Conditional 50.1% (40.2% to 57.6%) Participant-associated 28.6% (17.7% to 34.2%)
Time above 250 lx meEDI during wake	Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	Unit h Median 2.417 IQR 0.850 to 4.617 Participants 141 Days 737		FDR supported; adjusted p 0.007 Part R-squared 8.6% (5.2% to 18.7%)	Estimate Ratio per 1 h: 1.136 (1.073 to 1.202) FDR supported; adjusted p <0.001 Part R-squared 6.4% (2.0% to 12.7%)	Marginal 18.0% (12.9% to 29.1%) Conditional 50.1% (41.6% to 58.0%) Participant-associated 32.1% (20.7% to 36.7%)
Time below 10 lx meEDI before sleep	Low-light time before bed; limits evening melatonin suppression and circadian delay.	Unit h Median 1.883 IQR 1.025 to 2.583 Participants 139 Days 655		FDR supported; adjusted p 0.049 Part R-squared 5.4% (3.2% to 13.4%)	Estimate Difference per 1 h: -0.134 (-0.201 to -0.066) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.2% to 10.5%)	Marginal 7.6% (4.6% to 17.2%) Conditional 38.3% (31.8% to 48.5%) Participant-associated 30.7% (21.7% to 38.1%)
Time below 1 lx meEDI during sleep	Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	Unit h Median 7.133 IQR 5.967 to 8.250 Participants 141 Days 778		FDR not supported; adjusted p 0.058 Part R-squared 5.4% (3.2% to 14.2%)	Estimate Ratio per 1 h: 0.987 (0.968 to 1.006) FDR not supported; adjusted p 0.227 Part R-squared 0.6% (0.0% to 3.8%)	Marginal 8.0% (5.1% to 18.2%) Conditional 40.9% (32.8% to 48.5%) Participant-associated 32.9% (22.0% to 37.8%)
Longest period above 250 lx meEDI	Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	Unit h Median 0.633 IQR 0.283 to 1.204 Participants 141 Days 816		FDR not supported; adjusted p 0.398 Part R-squared 2.2% (1.4% to 8.1%)	Estimate Ratio per 1 h: 1.125 (1.065 to 1.187) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.3% to 9.6%)	Marginal 10.0% (6.1% to 17.9%) Conditional 36.3% (29.9% to 45.3%) Participant-associated 26.3% (18.7% to 33.0%)

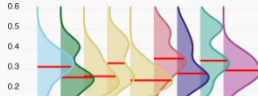
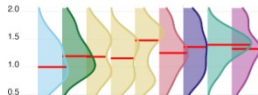
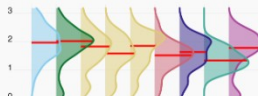
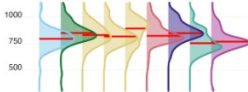
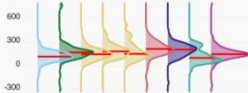
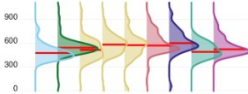
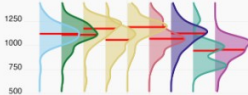
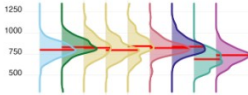
Near-eye personal light-exposure metrics and their geographic and photoperiod context						
Definition/relevance		Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Dynamics						
Interdaily stability	Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	Unit dimensionless Median 0.308 IQR 0.248 to 0.380 Participants 141 Days 816		FDR not supported; adjusted p 0.228 Part R-squared 6.9% (4.1% to 20.4%)	Estimate Odds ratio per 1 h: 0.983 (0.944 to 1.024) FDR not supported; adjusted p 0.401 Part R-squared 0.4% (0.0% to 4.6%)	Marginal 15.4% (10.4% to 30.8%) Conditional 15.4% (10.4% to 30.8%) Participant-associated Not available
Intradaily variability	Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	Unit dimensionless Median 1.253 IQR 0.930 to 1.502 Participants 141 Days 816		FDR not supported; adjusted p 0.483 Part R-squared 5.1% (3.8% to 18.7%)	Estimate Difference per 1 h: –0.019 (–0.056 to 0.019) FDR not supported; adjusted p 0.327 Part R-squared 0.7% (0.0% to 5.7%)	Marginal 5.9% (4.6% to 21.2%) Conditional 5.9% (4.6% to 21.2%) Participant-associated Not available
Exposure history						
meLEDI dose	Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	Unit klx·h Median 4.960 IQR 1.936 to 12.313 Participants 141 Days 761		FDR not supported; adjusted p 0.209 Part R-squared 2.8% (1.7% to 8.6%)	Estimate Ratio per 1 h: 1.275 (1.169 to 1.391) FDR supported; adjusted p <0.001 Part R-squared 7.5% (3.1% to 12.9%)	Marginal 11.8% (7.7% to 19.7%) Conditional 33.8% (26.7% to 42.8%) Participant-associated 22.0% (14.5% to 28.7%)

Table continued

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
		Descriptive summary		Association evidence		Modelled variation
	Definition/relevance	Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Level						
Mean melEDI	Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	Unit lx Median 5.154 IQR 2.831 to 9.225 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 8.4% (4.8% to 16.6%)	Estimate Ratio per 1 h: 1.181 (1.109 to 1.257) FDR supported; adjusted p <0.001 Part R-squared 7.7% (2.9% to 13.9%)	Marginal 25.4% (18.6% to 34.4%) Conditional 57.7% (51.9% to 65.1%) Participant-associated 32.3% (24.5% to 39.4%)
Brightest 10 h geometric mean	Offset geometric mean melEDI in the brightest supported 10-hour window; describes the strength of the main sustained daily exposure.	Unit lx Median 110.566 IQR 41.513 to 243.212 Participants 141 Days 816		FDR supported; adjusted p 0.011 Part R-squared 5.7% (3.3% to 12.7%)	Estimate Ratio per 1 h: 1.243 (1.133 to 1.363) FDR supported; adjusted p <0.001 Part R-squared 5.8% (1.9% to 11.1%)	Marginal 17.6% (12.1% to 26.0%) Conditional 46.0% (39.6% to 54.3%) Participant-associated 28.4% (20.9% to 35.4%)
Darkest 10 h geometric mean	Offset geometric mean melEDI in the darkest supported 10-hour window; describes the sustained low-light period.	Unit lx Median 0.103 IQR 0.020 to 0.253 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 13.7% (9.2% to 23.9%)	Estimate Ratio per 1 h: 1.088 (1.034 to 1.144) FDR supported; adjusted p 0.002 Part R-squared 3.4% (0.5% to 8.3%)	Marginal 22.0% (15.8% to 31.8%) Conditional 61.1% (55.3% to 68.1%) Participant-associated 39.1% (30.4% to 46.2%)
Spectrum						
Melanopic daylight efficacy ratio	Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	Unit dimensionless Median 0.724 IQR 0.643 to 0.795 Participants 137 Days 687		FDR supported; adjusted p <0.001 Part R-squared 9.6% (6.0% to 17.8%)	Estimate Difference per 1 h: 0.024 (0.017 to 0.031) FDR supported; adjusted p <0.001 Part R-squared 13.0% (6.7% to 20.6%)	Marginal 27.8% (21.3% to 38.1%) Conditional 60.5% (54.6% to 68.2%) Participant-associated 32.7% (24.1% to 40.0%)

Table continued

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
		Descriptive summary		Association evidence		Modelled variation
	Definition/relevance	Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Timing						
Midpoint of the brightest 10 hours	Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	Unit clock time Median 13:44 IQR 12:48 to 15:00 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 6.5% (4.4% to 12.6%)	Estimate Difference per 1 h: 0.054 (-0.045 to 0.153) FDR not supported; adjusted p 0.305 Part R-squared 0.2% (0.0% to 1.8%)	Marginal 6.7% (4.6% to 13.0%) Conditional 22.9% (16.9% to 31.6%) Participant-associated 16.2% (9.6% to 22.4%)
Midpoint of the darkest 10 hours	Centre time of the darkest 10 hours; indexes the main daily darkness cue.	Unit clock time Median 02:54 IQR 02:01 to 03:50 Participants 141 Days 816		FDR supported; adjusted p 0.020 Part R-squared 4.6% (3.0% to 10.5%)	Estimate Difference per 1 h: -0.148 (-0.245 to -0.050) FDR supported; adjusted p 0.004 Part R-squared 2.1% (0.3% to 5.6%)	Marginal 8.6% (5.9% to 15.8%) Conditional 28.2% (22.1% to 36.7%) Participant-associated 19.6% (12.6% to 25.8%)
First light timing above 250 lx meEDI	First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	Unit clock time Median 09:08 IQR 08:02 to 10:40 Participants 140 Days 727		FDR not supported; adjusted p 0.112 Part R-squared 3.8% (2.3% to 10.2%)	Estimate Difference per 1 h: -0.097 (-0.247 to 0.053) FDR not supported; adjusted p 0.227 Part R-squared 0.5% (0.0% to 2.8%)	Marginal 6.8% (4.4% to 14.4%) Conditional 31.0% (24.6% to 40.5%) Participant-associated 24.2% (16.2% to 31.5%)
Last light timing above 250 lx meEDI	Last bright-light exposure; later timing may delay circadian phase and sleep onset.	Unit clock time Median 18:08 IQR 16:27 to 19:42 Participants 141 Days 687		FDR supported; adjusted p <0.001 Part R-squared 11.6% (8.3% to 18.6%)	Estimate Difference per 1 h: 0.323 (0.186 to 0.460) FDR supported; adjusted p <0.001 Part R-squared 4.5% (1.5% to 8.5%)	Marginal 22.2% (17.0% to 30.2%) Conditional 37.9% (31.6% to 46.3%) Participant-associated 15.8% (9.7% to 22.3%)
Mean timing of exposure above 250 lx meEDI	Average bright-light timing; summarizes the phase of daily circadian stimulation.	Unit clock time Median 13:29 IQR 12:29 to 14:39 Participants 141 Days 742		FDR supported; adjusted p <0.001 Part R-squared 12.7% (9.4% to 19.2%)	Estimate Difference per 1 h: 0.121 (0.028 to 0.213) FDR supported; adjusted p 0.013 Part R-squared 1.2% (0.1% to 3.6%)	Marginal 14.7% (10.7% to 21.9%) Conditional 25.7% (19.8% to 34.2%) Participant-associated 11.0% (5.3% to 17.2%)
Overall distributions are median (interquartile range). Participants is the participant count and Days is the participant-day count. Site and photoperiod part-R ² values can contain overlapping fitted information and must not be summed. Participant-associated R ² is the conditional-minus-marginal contribution associated with the participant random intercept and is undefined for participant-level outcomes. The MDER uses the mean of viable minute-level ratios in both the descriptive summary and geographic-association model cells.						

265 Immediate environments and routines reorganise exposure

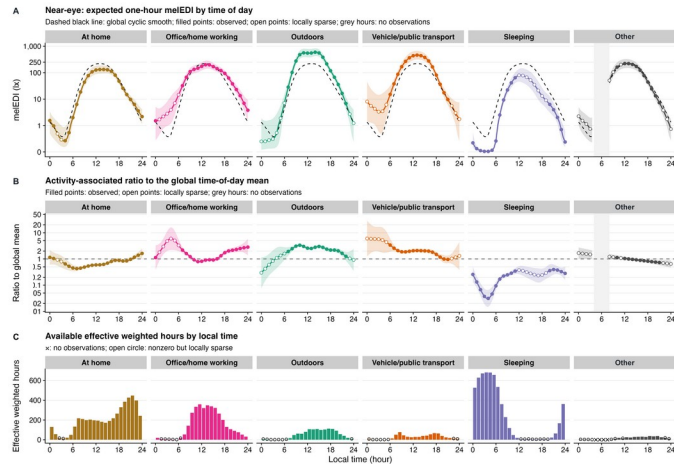
266 Self-reported light source was strongly associated with one-hour near-eye exposure, with an
267 FDR-supported light-source-by-site interaction. Site-average interaction-model estimates
268 were 959.6 lx melanopic EDI for outdoor daylight (95% CI 786.8 to 1,170.2), 198.8 lx for
269 indoor daylight (169.3 to 233.4), 85.1 lx for indoor electric light (71.6 to 101.1) and 23.8 lx for
270 a display screen (17.4 to 32.5). Outdoor daylight was 11.27 times the indoor-electric
271 estimate (8.71 to 14.60) and exceeded the 250-lx daytime recommendation, whereas the
272 indoor-daylight and indoor-electric intervals were below it. During reported sleep, darkness
273 and external-light estimates were 2.24 lx (1.54 to 3.26) and 11.18 lx (8.06 to 15.51), both
274 above the 1-lx bedside upper limit. Outdoor electric light remained imprecise relative to
275 indoor electric light near eye (ratio 1.019, 0.336 to 3.090; FDR-adjusted $p = 0.973$). These
276 contextual averages do not classify people or hours as adherent.

277 External light during sleep in Munich (DE) was estimated at 130.2 lx, 11.65 times the
278 corresponding site average; sleep darkness in Kumasi (GH) was 0.371 lx, 0.166 times its
279 site average. Sparse site-category cells limited some estimates.

280 The primary analysis used a population-mean quasi-Tweedie model with participant-cluster-
281 robust covariance for uncertainty; lag-one residual correlation remained 0.288. Fitted versus
282 observed exact-zero proportions were 39.8% and 27.8%, respectively. An exploratory
283 mixed-effects model quantified variance explained after adding a participant intercept to site-
284 by-light-source fixed terms. Marginal R^2 was 0.796 and conditional R^2 0.876, an 8.0-
285 percentage-point increase in explained variance from the participant intercept. Of the
286 marginal R^2 , light source accounted for 89.3%, site 7.1% and their interaction 3.7%.
287 Separately, an exploratory time-of-day model attributed 60.4% of variation in fitted hourly
288 patterns to global time of day and 24.2% to light-source deviations. Among these two
289 components, light source accounted for 28.6%. The remaining shares were 5.2% for site,
290 7.6% for participant patterns and 2.5% for day-to-day shifts.

291 A sensitivity adjusted the population-average light-source comparisons for each participant's
292 overall category distribution, distinguishing hour-to-hour differences within participants from
293 differences in category composition between participants. The overall category pattern was
294 retained. For external light during sleep, the near-eye ratio relative to indoor electric light
295 changed from 0.449 in the primary additive model to 0.333 (95% CI 0.190 to 0.585), a 25.8%
296 decrease. Thus, between-participant category composition influenced the magnitude, but not
297 the direction, of this contrast. [Supplementary Figure S8](#) shows the temporal, support and
298 site-specific results.

299 Immediate setting combined behaviour and environment, with an FDR-supported activity-by-
 300 site interaction. Site-average near-eye estimates were 714.2 lx outdoors (95% CI 561.7 to
 301 908.2), 331.2 lx on the road or in a vehicle (274.4 to 399.8), 198.9 lx during office or home
 302 work (173.9 to 227.5), 76.4 lx while awake at home (62.0 to 94.1) and 4.28 lx during sleep
 303 (3.06 to 5.99). The sleep category included many low or zero observations and occasional
 304 high stray-light readings, so its fitted mean need not represent a typical observation. Relative
 305 to awake time at home, the ratios were 9.35 outdoors, 4.34 for travel, 2.60 for work and
 306 0.056 during sleep. The outdoor estimate exceeded the 250-lx daytime recommendation, the
 307 work estimate remained below it, and the sleep estimate exceeded the 1-lx bedside-
 308 environment upper limit. Borås (SE) had a fitted outdoor estimate of 3,054 lx, 4.28 times the
 309 site average, whereas Tübingen (DE) had 127 lx, 0.18 times the site average. In an
 310 exploratory mixed model of five setting categories, marginal R^2 was 0.766 near eye and
 311 0.768 at the chest. Adding the participant intercept raised conditional R^2 to 0.858 and 0.859,
 312 an explained-variance increment of 9.2 percentage points at both positions. Of the marginal
 313 R^2 , immediate setting accounted for 80.7% near eye and 86.1% at the chest, site 12.5% and
 314 5.9%, and their interaction 6.9% and 8.0%, respectively. [Figure 3](#) shows the immediate-
 315 setting patterns and site-specific estimates. Separately, an exploratory near-eye time-of-day
 316 model attributed 52.5% of variation in fitted hourly patterns to time of day and 30.7% to
 317 immediate setting. Among these two components, immediate setting accounted for 36.9%.
 318 The remaining shares were 5.6% for site, 8.4% for participant patterns and 2.8% for day-to-
 319 day shifts.



Bands are conditional pointwise 95% intervals; no simultaneous-band or curve-wide inferential claim is made.
Sites are assessed equally on the fitted too-mean scale; participant and participant-day smooths are excluded from displayed curves.
Panel B divides each displayed activity curve by the displayed global time-of-day mean; the dashed reference is 1. Other is display-only.
The activity palette is distinct from the light-source category colours used elsewhere in this report series.
Each multi-select participant-hour contributes 1% to every related activity. Filled circles mark observed support, open circles mark locally sparse support.
Grey regions and x mark zero observations. Only the global smooth is cyclic; thin-plate activity and site deviations may separate over midnight.
No line is wrapped from 24:00 to 00:00. The melEDI axis transformation is display-only.

D Near-eye activity-by-site interaction model

Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

	At home P 128 · D 709 · H 5,106	Office/home working P 124 · D 511 · H 3,624	Outdoors P 121 · D 485 · H 1,422	Vehicle/public transport P 111 · D 388 · H 816	Sleeping P 126 · D 714 · H 3,906	Other P 72 · D 158 · H 381
Site-average estimate	76.4 lx (62.6–94.1)	198.9 lx (173.9–227.5)	714.2 lx (561.7–908.2)	331.2 lx (274.6–399.8)	4.3 lx (3.1–6.0)	221.8 lx (97.2–506.4)
Reference category	2.684x home (2.121–3.198)	9.354x home (6.846–12.595)	4.338x home (3.329–5.668)	0.056x home (0.041–0.076)	2.870x home (1.228–7.162)	Additive display only
FDR p (reference)	FDR p < 0.001	FDR p < 0.001	FDR p < 0.001	FDR p < 0.001	FDR p < 0.001	
• Borås (SE)	1.74x (0.94–3.22) FDR p 0.152	1.64x (1.22–2.18) FDR p 0.005	4.28x (2.84–6.45) FDR p < 0.001	1.67x (0.93–3.05) FDR p 0.159	0.45x (0.22–0.96) FDR p 0.084	Not estimable
• Delft (NL)	2.87x (1.38–3.28) FDR p 0.009	1.48x (1.67–2.84) FDR p 0.048	2.32x (1.32–4.09) FDR p 0.014	1.32x (0.96–2.02) FDR p 0.209	1.17x (0.51–2.68) FDR p 0.797	Not estimable
• Dortmund (DE)	2.31x (0.14–4.87) FDR p 0.051	1.67x (0.81–3.96) FDR p 0.169	2.80x (1.35–3.34) FDR p 0.025	2.47x (1.44–4.89) FDR p 0.005	4.53x (1.59–15.79) FDR p 0.048	Not estimable
• Tübingen (DE)	0.42x (0.28–0.63) FDR p < 0.001	0.54x (0.39–0.73) FDR p < 0.001	0.18x (0.11–0.30) FDR p < 0.001	0.34x (0.21–0.56) FDR p < 0.001	0.38x (0.16–0.89) FDR p 0.061	Not estimable
• München (DE)	1.25x (0.73–2.14) FDR p 0.486	1.54x (1.00–2.36) FDR p 0.102	1.34x (0.72–2.48) FDR p 0.433	0.81x (0.58–1.13) FDR p 0.299	10.47x (4.40–24.89) FDR p < 0.001	Not estimable
• Madrid (ES)	0.64x (0.46–0.89) FDR p 0.026	1.14x (0.93–1.41) FDR p 0.209	0.97x (0.47–1.98) FDR p 0.963	0.65x (0.33–1.30) FDR p 0.303	0.78x (0.20–3.10) FDR p 0.797	Not estimable
• Izmir (TR)	0.60x (0.56–1.16) FDR p 0.314	0.79x (0.59–1.07) FDR p 0.196	0.97x (0.47–1.97) FDR p 0.963	1.14x (0.96–1.96) FDR p 0.732	1.61x (0.99–2.89) FDR p 0.185	Not estimable
• San José (CR)	0.78x (0.46–1.31) FDR p 0.428	0.50x (0.33–0.76) FDR p 0.006	Not estimable	0.98x (0.53–1.84) FDR p 0.960	0.50x (0.21–1.21) FDR p 0.196	Not estimable
• Kumasi (GH)	0.57x (0.20–1.62) FDR p 0.378	0.67x (0.45–0.99) FDR p 0.088	0.52x (0.33–0.82) FDR p 0.019	0.93x (0.57–1.53) FDR p 0.639	0.16x (0.07–0.42) FDR p 0.001	Not estimable

Site-average estimates and comparisons for the five named categories use the activity-by-site interaction model.
Other is the additive display-only estimate and was excluded from the interaction model.
Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals.
Bold site ratios and FDR p-values pass the complete 44-member near-eye family; unsupported cells are not estimable.
Five-category interaction frame: 126 participants; 724 participant-days; 16,135 unique participant-hours; 16,875 generated long rows;
16,135 effective weighted hours; nine sites; 4,748 exact-zero participant-hours. Each multi-select hour contributes 1%.
Other-only display support: P 72 · D 159 · H 391 · P, participants; D, participant-days; H, unique participant-hours.

Figure 3: Personal light exposure by immediate setting across time of day and study sites. The diary variable labelled activity combines behaviour and environmental setting. **A–C**, Exploratory time-of-day analysis. Panel **A** shows site-average expected one-hour melanopic EDI with clock-specific 95% confidence intervals; the dashed black curve is the site-average local-clock smooth. Panel **B** shows each setting curve relative to that smooth, with 1 as the reference. Panel **C** shows effective weighted hours; open circles indicate locally sparse support and grey regions indicate times without observations. **D**, Site-average category estimates and site-specific deviation ratios from the activity-by-site interaction model, with participant-cluster-robust 95% confidence intervals and FDR-adjusted p values. Each multi-select participant-hour contributes weight $1/k$ to every retained setting category. The Other category is sparse and display-only. The fitted frame contains 126 participants, 724 participant-days, 16,135 unique participant-hours, 16,135 effective weighted hours and 4,746 exact-zero participant-hours. Estimates are observational.

Across models of free versus work day, active versus sedentary daily activity status and previous-night sleep duration, the site-average reference was a work, sedentary day following an average 7.8 h of sleep; expected near-eye exposure was 89 lx melanopic EDI (95% CI 67 to 119). Active versus sedentary status was associated with 2.06 times the expected exposure (1.54 to 2.76; FDR-adjusted $p < 0.001$). A model constraining the free-versus-work association across sites estimated 1.45 times higher exposure on free days (1.14 to 1.85; FDR-adjusted $p = 0.004$). Allowing the association to differ by site gave an inconclusive site-average ratio of 1.15 (0.93 to 1.44; FDR-adjusted $p = 0.298$), while some sites differed (interaction $p = 0.017$). Exploratory ratios were FDR-retained in Borås (SE; 1.95, 1.26 to 3.01) and Dortmund (DE; 2.44, 1.42 to 4.22), with FDR-adjusted $p = 0.013$ for each. The free-versus-work result was also inconclusive under the alternative working variance and sensitive to omitting either site. Each additional hour of previous-night sleep duration, centred at 8 h, had an inconclusive ratio of 0.98 (0.85 to 1.12; FDR-adjusted $p = 0.745$). Excluding six participants outside the employment requirement left 15,871 supported hours from 684 participant-days and 131 participants across all nine sites. The free-versus-work, active-versus-sedentary and previous-sleep associations remained stable within model uncertainty; the predictor-by-site interaction for day type also remained FDR-supported. [Supplementary Table S9](#) and Supplementary Figures [S9](#), [S10](#), [S11](#) and [S12](#) report common, temporal, position-specific and site-specific results.

355 Person-level correlates are selective

356 Self-report analyses did not support general exposure profiling. None of 68 prespecified
357 near-eye associations between four light-exposure-related behaviour (LEBA) factors and 17
358 metrics retained FDR support. Unsatisfactory model checks made the sleep-environment
359 metric unfit for inference, so its estimates were suppressed. Neither average associations
360 nor site interactions across nine visual-light-sensitivity (VLSQ-8) outcomes retained FDR
361 support at either position. Supplementary Tables [S10](#), [S11](#) and [S12](#), and Supplementary
362 Figures [S13](#) and [S14](#), give the full results.

363 Each hour later in corrected midsleep on free days (MCTQ questionnaire) was associated
364 with first exposure above 250 lx occurring 0.381 h later (95% CI 0.146 to 0.616; FDR-
365 adjusted $p = 0.003$). Each 10-point shift towards greater morning preference (MEQ
366 questionnaire) was separately associated with first exposure above 250 lx occurring 0.444 h
367 earlier (0.189 to 0.699 h earlier; FDR-adjusted $p = 0.001$). The brightest- and darkest-10-
368 hour midpoints followed the same directions; last exposure and the longest-bright-period
369 midpoint did not retain support. The distinct questionnaires were not combined.
370 [Supplementary Table S13](#) and [Supplementary Figure S15](#) show adjusted estimates and
371 observed timing distributions.

372 Each additional decade of age was associated with higher near-eye brightest-10-hour mean,
373 time above 1,000 lx and daily dose, with ratios from 1.27 to 1.31. These cross-sectional
374 associations may reflect cohort, occupation, behaviour or other confounding, not ageing
375 itself. Neither near-eye metric associations with biological sex nor near-eye age-by-site or
376 sex-by-site interactions met the FDR criterion. In a separate daily-pattern analysis, near-eye
377 and chest profiles differed by biological sex (FDR-adjusted $p = 0.028$ and 0.029), but the
378 difference did not meet the FDR criterion after restriction to records with complete
379 immediate-setting data and adjustment. This attenuation is compatible with immediate-
380 setting patterning, but restriction and adjustment cannot be disentangled as mechanisms.
381 Gender was recorded separately but not analysed; its category matched biological sex in all
382 but one case. Supplementary Tables [S10](#), [S14](#) and [S15](#), and Supplementary Figures [S16](#)
383 and [S17](#), give the complete results.

384 Measurement position and robustness constrain 385 interpretation

386 The paired sample of 112 participants and 643 participant-days allowed separate near-eye
387 and chest models to be compared within the same participants and participant-days. The full

388 chest dataset of 154 participants and 902 participant-days broadened coverage beyond the
389 141 participants and 816 participant-days near eye. At several sites, especially San José
390 (CR), offering a chest sensor enabled recruitment among people reluctant to wear the
391 glasses-mounted sensor. This improved feasibility, not the ocular-exposure sample.

392 A parallel analysis of overlapping MeLiDos data reported structured position differences
393 across scale, metric, context, site, day and participant³³. Our separate and common-sample
394 analyses likewise show directional agreement but differences in magnitude or support. Chest
395 adherence retained the main daytime and pre-sleep directions and interval conclusions, but
396 provides no sleep ocular claim. We neither pooled positions nor inferred equivalence or a
397 universal correction.

398 Non-estimable planned quantities were four sleep-environment variants (main plus three
399 sensitivity variants) at each position in the light-behaviour analysis (LEBA), and one longest-
400 period sensitivity comparison incompatible with the metric representation. We retained these
401 outcomes rather than replacing them with zero or null findings. Qualifications for zeros,
402 residuals, dependence, nonlinear-term overlap, sparse support and influence remain with
403 the affected analyses and in Methods.

404 Discussion

405 The expected 24-hour rhythm was evident across nine sites in seven countries. The key
406 advance was to quantify its amplitude across sites, participants and participant-days, and to
407 show how immediate settings reorganised exposure within that rhythm. Across analyses, the
408 shared local-clock pattern accounted for the largest share of full-model R^2 in the daily-pattern
409 analysis. Separate hourly decompositions showed large light-source and immediate-setting
410 contrasts, with the next-largest shares of variation in fitted hourly patterns after time of day.
411 Participant patterns and participant-day shifts together exceeded site patterns in the daily-
412 pattern model, while site, photoperiod and measured person-level associations were metric-
413 specific. Because these analyses used different outcomes and decompositions, this is a
414 qualitative synthesis rather than a single quantitative ranking.

415 The consensus recommendations provide the clearest health-relevant benchmark. Only
416 24.0% of waking minutes reached the daytime range, while 63.3% of pre-sleep and 87.7% of
417 bedside sleep-environment minutes met their upper limits. Modelled adherence was lower
418 on free days during daytime and sleep but higher during pre-sleep, with site variation in
419 daytime and sleep. Outdoor exposure exceeded the daytime recommendation, but all other
420 conditions remained below it on average. The fitted sleep-category mean exceeded the

421 bedside limit, while 87.7% of pooled bedside sleep-environment minutes met it. These
422 quantities describe different aspects of the exposure distribution and should be interpreted
423 together. These pooled fractions, recommendation-window proportions and hourly means
424 are distinct estimands. Together, they identify intervention opportunities and shortfalls. The
425 recommendations principally address healthy adults aged 18 to 55 with regular daytime
426 schedules⁸.

427 This baseline connects mechanistic recommendations with population evidence. Large wrist-
428 sensor cohorts have associated day and night light patterns with psychiatric disorders,
429 incident type 2 diabetes, cardiovascular disease and mortality, and light timing with sleep
430 timing and sleepiness¹⁰⁻¹⁵. Other cohorts have linked evening, nocturnal or strongly seasonal
431 light with sleep in bipolar disorder, obesity, diabetes, hypertension and metabolic health¹⁶⁻¹⁹.
432 These outcomes span sleep and major non-communicable diseases. The exposome frames
433 environmental contributions across the life course^{20,21} but rarely includes time-resolved
434 ocular light. This evidence supports personal-light measurement, not clinical risk
435 classification of our exposure categories. Most cohorts used wrist or bedside illuminance
436 without resolving melanopic exposure at the eye, immediate light source or its multiscale
437 architecture.

438 Exploratory inverse associations of average daytime with sleep and pre-sleep adherence
439 show why the three recommendation windows should not be collapsed into one score. They
440 may reflect schedules, environments or other between-participant differences, not a within-
441 person trade-off or stable phenotype. In exploratory nonlinear models, light source and
442 immediate setting accounted for substantially more variation in fitted hourly patterns than
443 participant-specific patterns: 24.2% versus 7.6%, and 30.7% versus 8.4%, respectively. This
444 finding is consistent with schedules and immediate environments being prominent correlates
445 of hourly exposure, but it is not a causal explanation for the cross-window adherence
446 association. Longitudinal models of dependence and changing routines are needed to test
447 whether improving one part of the pattern displaces or supports another.

448 Location mattered, but not as one north-to-south ranking. Site differences concentrated in
449 daily level, waking bright-light duration and several timing metrics, while dose, spectrum and
450 regularity did not show the same site pattern. Civil photoperiod was associated with a
451 broader set of metrics, with larger day-length associations for bright-light level and dose than
452 for the darkest daily window. Between 14.2 and 16.2 h of civil photoperiod, six of nine
453 metrics showed signs of uncertainty about a photoperiod effect beyond that point. Sites were
454 sampled in distinct calendar periods and covered unequal photoperiod ranges, leaving

455 latitude, season, weather, temperature, culture, mobility and the built environment entangled.
456 Repeated seasonal sampling across more sites is needed to separate these influences.

457 Immediate setting patterned exposure, consistent with environmental opportunity, built
458 settings and behaviour³⁶. Outdoor daylight and being outdoors exceeded indoor electric light
459 and awake time at home, respectively; active days exceeded sedentary days. Site
460 interactions placed outdoors-category exposure higher in Borås (SE) than Tübingen (DE),
461 while the free-versus-work association was most evident in Borås (SE) and Dortmund (DE).
462 Participant intercepts captured remaining between-participant variation, not individual
463 responses or predictions. Future monitoring should preserve local clock time, location,
464 immediate setting, daylight access and site; self-selection, occupation, mobility and weather
465 remain potential confounders.

466 Light-behaviour and visual-sensitivity scores did not meet their FDR criteria, whereas
467 chronotype aligned with selected timing outcomes and age with selected bright-light and
468 dose metrics. This accords with evidence linking circadian preference or timing to daylight
469 exposure, sleep and routine, without identifying causal direction^{37–40}. Metric-level biological-
470 sex comparisons were inconclusive near the eye. The complete available-data curve differed
471 by biological sex, but immediate-setting-complete analyses did not retain this separation.
472 Person-level patterns may concern specific exposure dimensions rather than a general high-
473 versus-low phenotype.

474 Near-eye sensing approximates the incident ocular field during wake but is not a retinal
475 measurement; chest sensing samples a related environmental field. Studies show structured
476 differences by body position, morphology, posture and surrounding light^{29–31}. The matched
477 sample and parallel placement study show metric- and context-specific MeLiDos differences
478 without a universal correction³³. The chest option broadened recruitment among people
479 unwilling to wear glasses-mounted sensors, especially in San José (CR). Study design must
480 balance biological proximity, burden and recruitment while keeping estimands position-
481 specific.

482 The harmonised protocol combined repeated near-eye sensing with local-clock and
483 contextual information across nine sites and seven countries. A generalised additive model
484 separated the shared 24-hour pattern from site, participant and participant-day layers
485 obscured by daily summaries. Collecting these data demanded substantial effort from
486 participants and study teams. Glasses-mounted loggers required fitting, explanation and
487 daily handling; removal, occlusion, charging, discomfort, weather and social acceptability all
488 affected measurement quality^{41–43}. Despite this burden, the protocol measured closer to the
489 incident ocular field at high temporal and contextual resolution.

490 Several limitations qualify these findings. Modest site samples limit site estimates,
491 interactions and subgroup evaluation. Healthy working-age adults may not represent
492 children, adolescents, older or clinical populations, shift workers, people outside
493 employment, or those with restricted mobility or atypical living conditions. Sparse latitude-by-
494 photoperiod coverage leaves latitude, seasonality, culture, climate and built environment
495 entangled and requires broader seasonal sampling. Retrospective hourly setting and light-
496 source reports enabled harmonisation but could not distinguish fine-grained settings or
497 behaviours within home, work or transport. Very low readings require caution. The
498 manufacturer specifies an operating range starting at 1 lx, and independent laboratory
499 testing began at 2 lx⁴⁴. A scene-based field validation found high overall agreement with
500 laboratory-grade spectral references, but underestimation increased at lower intensities and
501 varied with lighting, time, scene complexity and site⁴⁵. Single-digit values therefore
502 approximate low-light exposure rather than provide precise measurements. Finally, no health
503 outcome was measured.

504 This nine-site reference locates near-eye exposure relative to physiology-based
505 recommendations, shows that variation among people and days within sites matters, and
506 identifies temporal, contextual and positional information for future cohorts. Taken together,
507 everyday ocular light exposure is structured, measurable and strongly shaped by behaviour
508 and micro-environment, but cannot be reduced to site, latitude or photoperiod. Next steps
509 are broader sampling across photoperiods, latitudes and locations and prospective links to
510 sleep, mental, metabolic, cardiovascular and other outcomes. Causal health benefits or
511 policy effects require interventions and representative longitudinal designs. Potential
512 interventions through clinical advice, occupational health, building design, urban planning or
513 public-health guidance must address the contexts in which people receive light. Time-
514 resolved near-eye measurement therefore offers a practical route from circadian biology to
515 precision prevention and context-sensitive environmental health.

516 Methods

517 Study design and sites

518 MeLiDos is a prospective multicentre observational field study of personal light exposure
519 conducted with a harmonised protocol³⁵. Participants completed an eight-day ambulatory
520 protocol comprising continuous light logging, repeated smartphone assessments and daily
521 diaries under free-living conditions. At an in-person registration visit, participants provided
522 consent, completed baseline questionnaires, received study instructions and were fitted with

523 the loggers. Seven consecutive recording days separated registration and return visits; most
524 sites recorded from Monday to Monday, while Dortmund (DE) recorded from Tuesday to
525 Tuesday. If a return visit was delayed, participants continued until formal study termination
526 and device retrieval. Data were collected between August 2023 and October 2025 at Borås
527 (SE), Delft (NL), Dortmund (DE), Tübingen (DE), Munich (DE), Madrid (ES), Izmir (TR), San
528 José (CR) and Kumasi (GH). Sites were an opportunity sample of consortium partners and
529 were not selected to represent national populations. Questionnaire materials were
530 administered in the local language except in Tübingen (DE) and Kumasi (GH), where
531 English was used, and translated materials followed a harmonised translation, review,
532 adjudication, pretesting and documentation process⁴⁶.

533 **Participants, recruitment and ethics**

534 The registered target population comprised adults aged 18 to 65 years in full-time
535 employment or part-time employment above 80%. Exclusion criteria included diagnosed
536 psychiatric or sleep disorders, shift work in the preceding two months, regular tobacco or
537 recreational drug use, photosensitising medication, visual impairment incompatible with the
538 devices and residence outside the local study area during recording. The retained near-eye
539 sample included nine participants outside the registered age or employment criteria: one
540 participant older than 65 years, two recorded as not employed and seven recorded as
541 marginally employed; one participant met both the age and employment criteria for
542 exclusion. As a near-eye-only eligibility sensitivity, we refitted the additive and predictor-by-
543 site models after excluding participants recorded as not employed or marginally employed.
544 Students and trainees were retained, and age above 65 years was not an exclusion criterion
545 for this sensitivity. The target was at least 15 participants per site, based on prior power
546 calculations and allowing for up to one-third dropout or incomplete recording⁴⁷. Participants
547 were recruited through local advertisements and institutional mailing lists. At some sites,
548 especially San José (CR), reluctance to wear the glasses-mounted near-eye sensor limited
549 recruitment, and the chest option broadened recruitment possibilities.

550 All participants provided written informed consent. Approvals were obtained from the Ethics
551 Committee of the Technical University of Munich for Tübingen (DE; 2023-115-S-KK) and for
552 Munich (DE) and Dortmund (DE; 2024-118-S-SB); the Dortmund (DE) site at the Federal
553 Institute for Occupational Safety and Health (BAuA) operated under this TUM multicentre
554 approval. The other approvals were from the Swedish Ethical Review Authority for Borås
555 (SE; 2024-06792-01), the THUAS Ethics Committee for Delft (NL; positive review dated 29
556 May 2024), the Research Ethics Committee of Universidad San Pablo-CEU for Madrid (ES;

557 834/24/106), the Scientific Research and Publication Ethics Committee of the Faculty of
558 Science and Engineering at İzmir Institute of Technology for Izmir (TR; decision 5/1), the
559 Comité Ético Científico of Universidad de Costa Rica for San José (CR; CEC-279-2025),
560 and the Committee on Human Research, Publications and Ethics of Kwame Nkrumah
561 University of Science and Technology for Kumasi (GH; CHRPE/AP/644/24). The protocol
562 was first trialled with participants at the Tübingen (DE) site, whose feedback informed
563 procedural refinements before implementation at the remaining sites; further details are
564 reported in the published protocol³⁵.

565 **Personal light measurements**

566 Personal light was recorded every 10 s with spectrally sensitive ActLumus loggers (Condor
567 Instruments, São Paulo, Brazil). Eight visible-band channels were processed by the device
568 to generate photopic and alpha-opic quantities, including melanopic EDI⁶. The manufacturer
569 specified a factory-calibrated operating range of 1 to 100,000 lx and 10% precision at 1,000
570 lx. Independent testing against a criterion spectrometer across indoor LED and outdoor
571 daylight conditions covered 2 to 100,000 lx and reported mean photopic error of +3.3% (SD
572 13.4%) for ActLumus⁴⁴. A subsequent scene-based field validation found high overall
573 agreement with laboratory-grade spectral reference instruments, together with systematic
574 underestimation that varied with intensity and setting⁴⁵. Participants wore the near-eye
575 device centrally on non-prescription spectacle frames and, where available, a
576 complementary pendant-mounted device at the chest. The near-eye position measures light
577 closer to the eye but is not a direct retinal measurement. The chest position provided a
578 separate measurement of the local environmental light field. Paired common-sample
579 analyses evaluated how closely it tracked the near-eye measurement as a potential ocular
580 proxy. During reported sleep, devices were placed facing upward on a bedside surface near
581 head level, so both positions characterise the bedside sleep environment. Participants
582 recorded removals and placed devices in an opaque bag during waking non-wear^{43,48}.

583 The placement analysis from the same project used overlapping MeLiDos data and is cited
584 as related methodological evidence, not independent replication³³. No analysis pooled
585 sensor positions or estimated a universal conversion. Common-sample analyses retained
586 the same participants and participant-days or participant-hours at both positions and fitted
587 each position separately.

588 Contextual and person-level measures

589 Participants completed the core Consensus Sleep Diary after awakening⁴⁹, together with
590 evening reports and paper diaries recording light source and an activity-environment
591 category for each hour. Smartphone assessments of sleepiness and mood were scheduled
592 at 11:00, 14:00, 17:00 and 20:00, and evening questionnaires assessed wellbeing and
593 physical activity⁵⁰⁻⁵². Smartphone measures and baseline or end-of-study questionnaires
594 used MyCap integrated with REDCap^{53,54}. Participants photographed and uploaded the
595 paper hourly diaries each day.

596 The modified Harvard Light Exposure Assessment categories were indoor electric light,
597 outdoor electric light, indoor daylight, outdoor daylight including shade, an emissive display,
598 darkness during sleep and external light during sleep⁵⁵. The primary light-source analysis
599 used one retained category per participant-hour. The diary variable labelled activity
600 combined behaviour and environmental setting. Its categories distinguished sleep, awake
601 time at home, motorised travel, active travel, office or home work, outdoor work, outdoor
602 leisure and other activities. We therefore refer to this variable as immediate setting in the
603 narrative. For analysis, outdoor travel, outdoor work and outdoor leisure were combined as
604 Outdoors. Reports could contain concurrent categories; duplicates were removed and an
605 hour with k retained labels contributed weight $1/k$ to each, preserving total weight one. Day
606 type was coded as work or free day, daily activity status as sedentary or active, and previous
607 sleep duration was linked from the preceding sleep period.

608 Baseline measures recorded age, biological sex, gender, employment and chronotype; end-
609 of-study measures covered visual light sensitivity, the sleep environment and habitual light-
610 exposure behaviour. Chronotype was represented separately by corrected midsleep on free
611 days from the Munich Chronotype Questionnaire and the Morningness-Eveningness
612 Questionnaire, whose higher scores indicate greater morning preference^{56,57}. Visual
613 sensitivity used the VLSQ-8⁵⁸, sleep environment used the Assessment of Sleep
614 Environment⁵⁹, and four Light Exposure Behaviour Assessment (LEBA) factors represented
615 outdoor time, device use in bed, ambient light before bed and morning or daytime light use⁶⁰.
616 Biological sex and gender were recorded as separate variables. Models analysed biological
617 sex, coded Female or Male, but did not analyse gender; its recorded category matched
618 biological sex in all but one case.

619 Time, sleep/wake status and coverage preparation

620 Processing preserved both actual time in Coordinated Universal Time and local wall-clock
621 time. Actual time ordered observations and measured elapsed gaps; local time aligned daily
622 patterns across sites. Repeated local minutes during autumn clock changes remained
623 distinct in actual time, and structurally absent spring minutes remained missing. Melanopic
624 EDI values at or above the validated 100,000-lx operating boundary were unavailable
625 analytically but retained in the raw channel with a flag.

626 Diary-defined sleep began at reported sleep preparation and ended at reported wake. Sleep
627 values were retained as the intended bedside record. Declared waking non-wear made the
628 analytical light value unavailable; missing wear labels were not treated as non-wear.
629 Complete 1,440-minute local participant-day grids preserved unsupported periods as
630 missing rather than deleting rows or replacing them with zero. A day was eligible when at
631 least 80% of this combined waking and bedside-sleep record had valid melanopic EDI.
632 Hourly summaries required at least 50% valid minutes in that hour, but an unsupported hour
633 did not erase valid minutes from the daily decision or minute-based metrics. Two otherwise
634 eligible near-eye days and three chest days with finite but entirely zero melanopic EDI were
635 excluded by the defined exact-all-zero screen.

636 Exposure metrics

637 Metrics were derived in R⁶¹ with LightLogR^{62,63}. The 17 outcomes represented
638 complementary properties of the daily pattern. Three level metrics described the daily
639 geometric mean melanopic EDI and the mean melanopic EDI in the brightest and darkest
640 10-hour windows. Five duration metrics quantified time above 1,000 lx, waking time above
641 250 lx, pre-sleep time below 10 lx, sleep-environment time below 1 lx and the longest
642 continuous period above 250 lx. Five timing metrics located the brightest and darkest
643 windows and the first, last and mean timing above 250 lx. Melanopic EDI dose integrated
644 exposure over time, melanopic daylight efficacy ratio characterised spectral composition
645 relative to visual illuminance, and interdaily stability and intradaily variability characterised
646 regularity and fragmentation. Geometric means were calculated from $\log_{10}(\text{melanopic EDI} +$
647 $0.1 \text{ lx})$, followed by inverse transformation and removal of the same offset⁶⁴. All-observed-
648 zero windows were retained as exact zero within verified numerical tolerance. Metric-specific
649 support rules made only the affected metric unavailable, not the participant-day. A longest-
650 period duration remained a lower bound when missing data could conceal a longer period;
651 exactly identified periods were examined separately.

652 The melanopic daylight efficacy ratio was the arithmetic mean of viable one-minute
653 melanopic EDI to photopic-illuminance ratios on a complete 1,440-position local-clock grid.
654 Both channels had to be finite and strictly positive, and at least 720 positions were required.
655 Repeated autumn minutes were averaged within each channel before division, and absent
656 spring minutes remained missing.

657 **Recommendation-window and adherence analyses**

658 We compared valid one-minute values with the Brown et al. recommendation ranges⁸: at
659 least 250 lx melanopic EDI during daytime, no more than 10 lx during pre-sleep and no more
660 than 1 lx during sleep. Daytime comprised diary-defined wake excluding the three hours
661 before reported sleep, pre-sleep comprised those three hours and sleep comprised the
662 reported sleep interval. Each descriptive numerator was divided by all valid near-eye-record
663 minutes in the corresponding window and pooled across the analysed sample. Daytime and
664 pre-sleep used the worn near-eye record; sleep used the bedside sleep-environment record.

665 The main adherence analysis retained one row per window period and defined adherence as
666 adherent valid minutes divided by all valid minutes in that period. A behavioural cycle
667 crossed midnight when necessary and inherited the work or free label of the calendar date
668 on which daytime began. The any-valid primary sample retained a window with at least one
669 valid comparison; an otherwise identical coverage sensitivity required at least 80% coverage
670 of the exact window interval. Exact 0% and 100% periods remained in an endpoint-inflated
671 beta-binomial model. Fixed predictors represented the complete window-by-site-by-day-type
672 pattern, and the retained random effect was a participant intercept in the beta-binomial
673 mean. Population averages weighted all nine sites equally. Separate FDR families covered
674 window-specific free-minus-work contrasts, interaction tests, site-specific contrasts against
675 zero and 27 comparisons of site-specific contrasts with their window-specific equal-site
676 values. Chest models were fitted separately for daytime and pre-sleep.

677 The exploratory cross-window extension used daytime adherence as the predictor and sleep
678 or pre-sleep adherence as outcomes in daytime-anchored behavioural cycles. It separated
679 each participant's cycle-specific daytime deviation from their observed average daytime
680 adherence, while adjusting for window, site, day type, their corresponding interactions and
681 the participant's mix of daytime observations across day types. The four within-participant
682 and between-participant associations formed one FDR family. Positive serial dependence
683 remained unresolved and some unchanged sensitivity fits were non-estimable, so no within-
684 participant day-level claim was stated. Descriptive profiles used site-average participant

685 means for 139 complete three-window profiles and 417 participant-window points; they
686 neither identify nor rank participants and do not define stable participant types.

687 **Daily temporal architecture**

688 The daily-pattern analysis used supported 30-minute arithmetic means, retaining a bin when
689 at least 15 one-minute values were valid. The outcome was $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. A
690 generalised additive model fitted with `mgcv::bam()` and `fREML` combined a cyclic common
691 local-clock smooth, sum-to-zero site curves, participant curves and participant-day random
692 intercepts^{65–68}. A boundary-aware first-order autoregressive working structure prevented
693 residual sequences from crossing participant-day, clock-repeat or gap boundaries. Fitted-
694 curve dispersion was compared on a common 48-bin grid. A Shapley decomposition
695 averaged each component's addition to R^2 across all component orders, thereby distributing
696 the model's explained variance^{69,70}. The decomposition partitioned the full model's row-
697 weighted in-sample R^2 for $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$, defined as the reduction in squared
698 prediction error relative to predicting the fitted-sample mean. The percentages therefore
699 partition full-model R^2 on that transformed response scale. They are not percentages of
700 variance explained in raw melanopic EDI and do not indicate causal or out-of-sample
701 predictive importance. Confidence intervals for dispersion and R^2 -share ratios used 2,000
702 hierarchical cluster resamples while holding the fitted model structure fixed.

703 **Geographic and photoperiod analyses**

704 We analysed 17 participant-level or participant-day metrics with metric-specific Gaussian or
705 Tweedie exponential-dispersion models^{71,72}. We treated overall site differences, civil
706 photoperiod, absolute latitude and the adequacy of a linear latitude term relative to site as
707 four separate model questions, each adjusted as a complete 17-test FDR family. Participant-
708 day models included participant random intercepts. Mixed-effects models were fitted with
709 `lme4` or `glmmTMB`, and component R^2 values were obtained with `performance` where
710 reported^{73–75}. Site and latitude were not entered as independent fixed predictors in one model
711 because each site has one latitude; site and latitude models instead used the same
712 observations and photoperiod adjustment. Site follow-up contrasts were calculated with
713 `emmeans` only after a supported overall site test and compared each site with a site-average
714 estimate, obtained by giving each included site equal weight⁷⁶.

715 Nonlinear photoperiod analyses fitted each of nine planned metrics separately with a
716 generalised additive model estimated by restricted maximum likelihood, a thin-plate
717 photoperiod smooth with basis dimension six and participant random effects nested within

718 site. The derivative, meaning the slope of the fitted curve, was evaluated with `gratia` at 100
719 equally spaced photoperiod values with 95% intervals⁷¹. A qualifying descriptive transition
720 required the preceding slope interval to be wholly above zero, the next interval to include
721 zero and every later interval through the recorded maximum to include zero as well. This
722 procedure located a change in the fitted slope but did not test equivalence, a mechanism or
723 a physiological ceiling. Expanded-basis, fixed-site, common-sample and leave-one-site-out
724 analyses assessed stability. The gap-timing-unaware sensitivity dataset still passed the
725 50%-per-hour and 80%-per-day rules, but did not use the timing of remaining missing
726 observations for an additional metric-specific adjustment.

727 Immediate-context and routine analyses

728 Light-source and activity-environment analyses modelled the zero-aware one-hour
729 geometric mean melanopic EDI with fixed-power quasi-Tweedie log-mean models and
730 participant-cluster-robust inference^{71,72}. Additive fixed-site models supplied primary category
731 tests. Category-by-site interaction models supplied site-average category estimates, ratios
732 and site-specific context when the interaction was supported. The activity-environment
733 analysis retained fractional $1/k$ weights for concurrent labels. FDR adjustments were applied
734 to the declared category and site-contrast families. Exact-zero calibration, residual
735 dependence, sparse cells and deletion refits were retained as model checks. A secondary
736 population-mean sensitivity added six participant-level category-proportion terms to the light-
737 source model while retaining participant-clustered covariance. This separated hour-to-hour
738 category differences within participants from between-participant differences in category
739 composition; it did not estimate participant random slopes. Exploratory `glmmTMB` Tweedie
740 log-link models fitted by maximum likelihood then added one common participant intercept to
741 site-by-category fixed effects at the near-eye position for light source and separately at both
742 positions for the activity-environment variable, with the power fixed at 1.539919^{74,75}. The
743 activity-environment models retained the five named categories, excluded Other-only hours
744 and used exact $1/k$ weights. Their fixed-predictor variance used the same fractional weights
745 so every participant-hour contributed one unit. Marginal and conditional R^2 used a lognormal
746 distribution-specific variance approximation, and hierarchy-respecting Shapley allocations
747 distributed marginal R^2 across site, category and their interaction. Percentages from the
748 exploratory time-of-day models instead partition variance among fitted linear-predictor
749 values. Because they do not compare this with observed outcome variance, they are not R^2 .
750 Results describe them as variation in fitted hourly patterns, not variance explained in raw
751 exposure. These were model-based point descriptions without bootstrap intervals. No model
752 included random category slopes.

753 The participant-hour analysis modelled each supported participant-hour with a zero-inclusive
754 quasi-Poisson log-link model, fixed site associations and participant-cluster HC3 uncertainty.
755 Free versus work day, active versus sedentary daily activity status and previous-night sleep
756 duration centred at 8 h were entered together. Three-test FDR families were applied
757 separately to pooled associations, predictor-by-site interactions and practical contrasts.

758 **Person-level analyses**

759 Light-related behaviour, visual sensitivity, chronotype, age and measured biological sex
760 were evaluated with site-adjusted metric-specific models. Participant-day outcomes
761 generally used Gaussian or Tweedie mixed models with participant random intercepts;
762 participant-level dynamics outcomes used fixed-effect models. Each analysis retained its
763 complete prespecified false-discovery-rate families, including non-estimable outcomes.
764 Chronotype instruments were modelled separately across five timing outcomes. Age
765 associations were scaled per decade, and female-minus-male contrasts used male as the
766 reference. Average associations and predictor-by-site interactions were distinct questions.

767 The complete biological-sex curve analysis used supported 30-minute arithmetic means
768 transformed as $\log_{10}(\text{melanopic EDI} + 0.1 \text{ lx})$. The generalised additive model included
769 cyclic common and female-specific curves, site curves, participant curves, participant-day
770 random effects and boundary-aware first-order autoregression. The primary test jointly
771 assessed the time-constant and time-varying female-minus-male contributions across the
772 complete 24-hour curve. Clock-specific uncertainty was participant-cluster robust. Level-
773 versus-shape attribution and adjustment for the activity-environment variable were
774 secondary or exploratory analyses.

775 **Multiplicity, uncertainty, model checks and sensitivities**

776 Tests were two-sided. We controlled the false discovery rate with the Benjamini-Hochberg
777 procedure separately within each declared complete family⁷⁸. We report 95% confidence
778 intervals and FDR-adjusted p values for confirmatory families. A result that did not meet an
779 FDR-adjusted criterion was treated as inconclusive rather than as evidence of absence or
780 equivalence. Unless stated otherwise, site-average estimates gave each included site equal
781 weight on the fitted scale before back-transformation. Exploratory analyses retained
782 separate multiplicity families and are labelled as such.

783 Model checks addressed convergence, rank, singularity, residual distribution, zero mass,
784 fitted support, heteroscedasticity, serial dependence, nonlinear-term overlap and participant
785 or site influence. Qualifications were attached to the affected results. Prespecified or

786 structured sensitivities included the separately prepared gap-timing-unaware dataset,
787 common sensor-position samples, alternative support or metric definitions, model-form
788 changes, participant deletion and leave-one-site-out checks. Near-eye and chest models
789 remained separate in every comparison. Analyses were conducted in R, with tidyverse
790 packages supporting data handling and reporting^{61,79}. Exact package versions, formulas,
791 inputs and output files are documented in the Supplementary Information and reproducibility
792 records.

793 Preregistration and deviations

794 The study was preregistered with AsPredicted (#273407)⁸⁰. The final analyses preserve the
795 registered scientific questions but include documented changes to placement priority, state
796 and coverage rules, metric definitions, outcome construction, model families, site structure,
797 multiplicity and sensitivity handling. The main manuscript reports the resulting estimands
798 rather than the historical hypothesis sequence. A complete mapping in the Supplementary
799 Information distinguishes scientific deviations and qualifications from analytical history and
800 technical provenance.

801 Artificial-intelligence assistance

802 During the original manuscript preparation, generative artificial intelligence tools (ChatGPT
803 5.5) were used to support language editing, structural refinement and text condensation. AI
804 assistance was used to improve clarity, coherence, concision and journal fit, but **not** to
805 generate, analyse or interpret primary data.

806 During primary data analysis, AI assistance was used to debug code sections, support figure
807 and table styling, and resolve modelling convergence and singularity issues. Participant
808 comments in their native languages were translated into English with AI assistance. All
809 translations were reviewed and approved or corrected by the respective site authors. AI
810 assistance was **not** used to develop the primary analysis strategy, write analysis scripts or
811 interpret results.

812 During review, an agentic AI assistant (ChatGPT 5.6 Sol) was used to review the analyses
813 for potential issues. This review prompted additional sensitivity analyses to evaluate data-
814 preparation and model-selection choices, expanded model diagnostics, and improved effect-
815 size reporting through Shapley decompositions and more precise language. AI assistance
816 also supported expanded analysis documentation and reproducibility checks.

817 All AI-assisted text was critically reviewed, edited, and approved by the authors, who take
818 full responsibility for the content, accuracy, integrity, and conclusions of the manuscript.

819 Data availability

820 All data used in this study are available through the [MeLiDos Project GitHub organisation](#)
821 and are archived in the [MeLiDos Zenodo community](#). Public site datasets have persistent
822 records for Kumasi (GH), Madrid (ES), Dortmund (DE), Izmir (TR), Tübingen (DE), San José
823 (CR), Munich (DE), Borås (SE) and Delft (NL)^{81–89}. These records and other project
824 resources are indexed through the MeLiDos Project Data Hub⁹⁰ and accessed reproducibly
825 through the documented `melidosData` interface⁹¹.

826 Code availability

827 All analysis code is available in the [project repository](#), with rendered reports on [GitHub](#)
828 [Pages](#) and an archive on [Zenodo](#). The archived version will be reconciled with the final
829 manuscript snapshot before submission.

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- 1075 • **Conceptualization:** J.Z., A.D., S.A., J.B., O.S. and M.S.
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1086 J.B., A.S.S., K.B., A.J. and M.S.
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- 1091 • **Visualization:** J.Z.
- 1092 • **Writing – original draft:** J.Z.
- 1093 • **Writing – review & editing:** all authors.

1094 Competing interests

1095 M.S. declares the following potential competing interests between 2021 and 2025: academic
1096 roles as a member of the Board of Directors of the Society of Light, Rhythms, and Circadian
1097 Health, Chair of Joint Technical Committee 20 of the International Commission on
1098 Illumination, member of the Daylight Academy, and Chair of the Research Data Alliance
1099 Working Group Optical Radiation and Visual Experience Data; remunerated roles as
1100 Speaker of the Steering Committee of the Daylight Academy, ad hoc reviewer for the Health
1101 and Digital Executive Agency of the European Commission and the Swedish Research
1102 Council, Associate Editor for *LEUKOS*, examiner for the University of Manchester, Flinders
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1111 the Daimler und Benz Stiftung; and being named on European Patent Application
1112 EP23159999.4A, “System and method for corneal-plane physiologically-relevant light
1113 logging with an application to personalized light interventions related to health and well-
1114 being.” M.S. declares that these roles and relationships had no influence on the work
1115 presented here.

1116 J.Z. declares the following potential competing interests between 2021 and 2025: academic
1117 roles as a member of Joint Technical Committee 20 of the International Commission on
1118 Illumination, member of the Research Data Alliance Working Group Optical Radiation and
1119 Visual Experience Data, and Speaker of group 2, melanopic effects of light, of the Technical
1120 Scientific Committee of the German Society of Lighting Technology and Design;
1121 remunerated roles as examiner for the Swiss Lighting Society; teacher for the German
1122 Society of Lighting Technology and Design, Munich University of Applied Sciences and the
1123 Technical University of Applied Sciences Rosenheim; associated partner at 3lpi lighting
1124 design + engineering, Munich; tool and three-dimensional-model designer for Zumtobel
1125 Lighting GmbH; and course designer for Munich University of Applied Sciences and Virtual
1126 University Bavaria; honoraria for talks from the German Society of Lighting Technology and
1127 Design, Lamilux/Heinrich Strunz GmbH, Robert Bosch Hospital Stuttgart, Ergotopia GmbH,
1128 the German Social Accident Insurance Institution for the Administrative Sector, BRIXEN
1129 CULTUR, KITEO GmbH & Co. KG and Augsburg University of Applied Sciences; travel
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1131 European Union Intellectual Property Office design patent 008194021-0001 through -0006
1132 for a non-visually optimized luminaire.

1133 A.D. declares academic roles as a member of Joint Technical Committee 20 of the
1134 International Commission on Illumination and Division Reporter DR6-50 for the “4th
1135 Manchester Workshop on Light Metrics for Biology: Light Pollution.” A.D. declares that these
1136 roles had no influence on the work presented here. D.B. is also owner of Minus 15 B.V., a
1137 company active in the lighting sector. All other authors declare no competing interests.

1138 Supplementary Information

1139 **Study support and exposure definitions**

1140 **Supplementary Table S1. Descriptive sample flow**

Descriptive sample flow and analytical support

Stage	Quantity	Count
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Participant roster

Available normalized participant metadata	Participants	191
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Primary near-eye sensor position

At least 80% complete before all-zero screen	Participant-days	818
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Exact all-zero days excluded	Participant-days	2
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Final descriptive dataset	Participants	141
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Final descriptive dataset	Participant-days	816
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Final descriptive dataset	One-minute real observations	1,175,160
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Complementary chest sensor position

At least 80% complete before all-zero screen	Participant-days	905
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Exact all-zero days excluded	Participant-days	3
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Final descriptive dataset	Participants	154
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Final descriptive dataset	Participant-days	902
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Final descriptive dataset	One-minute real observations	1,298,880
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Paired main subset

Paired main subset	Participants	112
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Paired main subset	Paired participant-days	643
--------------------	-------------------------	-----

Counts use their stated denominator. An excluded participant-day is an all-zero day, not necessarily an excluded participant. The paired row reports paired participant-days, not minute-level observation pairs.

1142 **Supplementary Table S1.** Participant, participant-day, and one-minute observation counts across the roster, sensor-specific completeness and all-zero
1143 screens, final descriptive datasets, and paired main subset. Each row names its denominator explicitly. An excluded participant-day is not necessarily an
1144 excluded participant.

1145 Supplementary Table S2. Full near-eye metric dictionary and distributions

Metric descriptive summary (near eye)							
Metric [†]	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)
Duration							
Time above 1,000 lx melEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) <i>01:10 ± 01:28</i> N=141; d=816	01:18 (00:31, 03:24) <i>02:00 ± 01:54</i> N=13; d=78	01:18 (00:29, 02:12) <i>01:33 ± 01:28</i> N=13; d=78	01:03 (00:20, 02:32) <i>01:48 ± 02:14</i> N=18; d=107	00:40 (00:07, 01:25) <i>01:06 ± 01:22</i> N=26; d=150	00:39 (00:10, 01:46) <i>01:09 ± 01:18</i> N=10; d=60
Time above 250 lx melEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) <i>03:00 ± 02:29</i> N=141; d=737	04:06 (02:30, 06:15) <i>04:21 ± 02:24</i> N=13; d=73	03:29 (02:03, 05:16) <i>03:39 ± 02:25</i> N=13; d=64	03:42 (02:05, 05:58) <i>04:05 ± 02:51</i> N=18; d=90	02:01 (00:47, 04:20) <i>02:52 ± 02:39</i> N=26; d=141	02:38 (00:55, 04:19) <i>02:57 ± 02:19</i> N=10; d=54
Time below 10 lx melEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) <i>01:51 ± 01:02</i> N=139; d=655	02:16 (01:30, 02:39) <i>02:05 ± 00:53</i> N=13; d=67	01:50 (01:21, 02:52) <i>01:59 ± 00:57</i> N=12; d=50	01:42 (00:53, 02:17) <i>01:41 ± 00:58</i> N=18; d=84	01:52 (01:04, 02:25) <i>01:52 ± 01:06</i> N=26; d=132	01:43 (00:42, 02:32) <i>01:45 ± 01:10</i> N=10; d=52
Time below 1 lx melEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) <i>07:04 ± 01:57</i> N=141; d=778	07:08 (06:03, 08:18) <i>07:12 ± 01:46</i> N=13; d=76	07:14 (05:56, 08:06) <i>07:01 ± 01:50</i> N=13; d=68	07:08 (05:56, 08:16) <i>07:05 ± 01:50</i> N=18; d=95	06:54 (05:51, 07:46) <i>06:47 ± 01:30</i> N=26; d=150	05:52 (04:41, 07:03) <i>05:58 ± 01:58</i> N=10; d=60
Longest period above 250 lx melEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) <i>00:55 ± 01:00</i> N=141; d=816	00:57 (00:28, 01:27) <i>01:12 ± 01:02</i> N=13; d=78	00:48 (00:23, 01:20) <i>01:09 ± 01:15</i> N=13; d=78	01:00 (00:21, 01:40) <i>01:20 ± 01:31</i> N=18; d=107	00:35 (00:13, 01:06) <i>00:50 ± 00:54</i> N=26; d=150	00:38 (00:17, 01:32) <i>01:01 ± 01:02</i> N=10; d=60
Dynamics							
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) <i>0.318 ± 0.095</i> N=141; d=816	0.311 (0.272, 0.367) <i>0.315 ± 0.078</i> N=13; d=78	0.258 (0.239, 0.389) <i>0.288 ± 0.094</i> N=13; d=78	0.264 (0.219, 0.319) <i>0.274 ± 0.074</i> N=18; d=107	0.329 (0.263, 0.424) <i>0.344 ± 0.107</i> N=26; d=150	0.244 (0.218, 0.277) <i>0.252 ± 0.058</i> N=10; d=60
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) <i>1.229 ± 0.389</i> N=141; d=816	0.997 (0.818, 1.388) <i>1.088 ± 0.343</i> N=13; d=78	1.193 (0.96, 1.389) <i>1.238 ± 0.345</i> N=13; d=78	1.182 (0.928, 1.49) <i>1.177 ± 0.407</i> N=18; d=107	1.154 (0.801, 1.442) <i>1.125 ± 0.354</i> N=26; d=150	1.48 (0.869, 1.642) <i>1.332 ± 0.477</i> N=10; d=60
Exposure history							
melEDI dose Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) <i>10.358 ± 16.216</i> N=141; d=761	9.296 (3.892, 21.895) <i>19.923 ± 27.784</i> N=13; d=71	10.495 (3.855, 17.395) <i>15.881 ± 20.395</i> N=13; d=74	6.751 (2.469, 14.435) <i>13.809 ± 23.711</i> N=18; d=97	3.862 (1.411, 10.217) <i>7.719 ± 9.847</i> N=26; d=137	7.11 (1.897, 15.789) <i>10.055 ± 11.237</i> N=10; d=55

1146

Table continued, column panel A

Metric descriptive summary (near eye)							
Metric [†]	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)
Level							
Mean melEDI Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	7.569 (4.55, 11.613) <i>9.171 ± 7.111</i> N=13; d=78	6.659 (4.18, 11.674) <i>8.274 ± 6.558</i> N=13; d=78	7.704 (3.081, 13.924) <i>12.017 ± 18.865</i> N=18; d=107	5.232 (3.224, 8.188) <i>7.221 ± 6.33</i> N=26; d=150	9.013 (4.011, 15.089) <i>12.102 ± 12.013</i> N=10; d=60
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	252.689 (104.514, 461.953) <i>399.302 ± 518.102</i> N=13; d=78	175.974 (78.625, 299.106) <i>262.206 ± 329.201</i> N=13; d=78	156.184 (67.783, 339.731) <i>422.54 ± 1,209.616</i> N=18; d=107	100.057 (33.255, 209.904) <i>179.372 ± 241.53</i> N=26; d=150	130.78 (54.044, 259.726) <i>201.947 ± 194.042</i> N=10; d=60
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.08 (0.016, 0.191) <i>0.183 ± 0.299</i> N=13; d=78	0.1 (0.035, 0.23) <i>0.2 ± 0.332</i> N=13; d=78	0.14 (0.036, 0.282) <i>0.223 ± 0.299</i> N=18; d=107	0.149 (0.074, 0.309) <i>0.273 ± 0.465</i> N=26; d=150	0.325 (0.119, 0.71) <i>0.515 ± 0.653</i> N=10; d=60
Spectrum							
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) <i>0.724 ± 0.116</i> N=137; d=687	0.788 (0.725, 0.872) <i>0.796 ± 0.162</i> N=13; d=74	0.743 (0.699, 0.796) <i>0.743 ± 0.077</i> N=13; d=65	0.781 (0.675, 0.853) <i>0.768 ± 0.118</i> N=18; d=97	0.65 (0.592, 0.728) <i>0.667 ± 0.098</i> N=26; d=144	0.724 (0.652, 0.773) <i>0.72 ± 0.093</i> N=10; d=55

Table continued, column panel A

Metric descriptive summary (near eye)							
Metric ¹	Unit	Overall	Borås (SE)	Delft (NL)	Dortmund (DE)	Tübingen (DE)	Munich (DE)
Timing							
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) <i>13:56 ± 01:51</i> N=141; d=816	13:16 (12:28, 14:10) <i>13:20 ± 01:33</i> N=13; d=78	14:12 (13:23, 15:18) <i>14:22 ± 01:51</i> N=13; d=78	13:52 (12:54, 15:04) <i>14:00 ± 01:42</i> N=18; d=107	13:38 (12:47, 14:44) <i>13:52 ± 01:42</i> N=26; d=150	14:54 (12:50, 16:03) <i>14:31 ± 02:14</i> N=10; d=60
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) <i>02:58 ± 01:41</i> N=141; d=816	02:02 (01:25, 02:48) <i>02:03 ± 01:07</i> N=13; d=78	02:55 (02:01, 03:27) <i>02:50 ± 02:05</i> N=13; d=78	02:32 (01:45, 03:22) <i>02:37 ± 01:19</i> N=18; d=107	03:08 (02:11, 04:10) <i>03:16 ± 01:34</i> N=26; d=150	02:37 (01:54, 04:17) <i>03:10 ± 02:00</i> N=10; d=60
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) <i>09:26 ± 02:22</i> N=140; d=727	07:57 (06:56, 08:33) <i>08:03 ± 01:32</i> N=13; d=74	09:04 (08:18, 09:52) <i>09:17 ± 01:56</i> N=13; d=68	08:36 (07:10, 10:13) <i>08:45 ± 02:21</i> N=18; d=93	09:43 (08:13, 10:53) <i>09:45 ± 02:21</i> N=25; d=133	09:36 (07:58, 10:53) <i>09:39 ± 03:12</i> N=10; d=56
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) <i>18:03 ± 02:36</i> N=141; d=687	18:41 (17:38, 19:38) <i>18:39 ± 01:38</i> N=13; d=73	18:30 (17:37, 19:59) <i>18:37 ± 02:08</i> N=13; d=62	19:34 (18:04, 20:42) <i>19:16 ± 02:28</i> N=18; d=89	17:37 (15:35, 19:09) <i>17:18 ± 02:44</i> N=26; d=121	19:50 (18:37, 20:50) <i>19:33 ± 02:11</i> N=10; d=53
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) <i>13:31 ± 01:52</i> N=141; d=742	13:20 (12:29, 14:24) <i>13:21 ± 01:25</i> N=13; d=72	13:52 (13:13, 15:06) <i>14:02 ± 01:41</i> N=13; d=71	13:44 (12:28, 14:55) <i>13:51 ± 01:56</i> N=18; d=93	13:17 (12:17, 14:31) <i>13:24 ± 01:45</i> N=26; d=138	14:04 (12:58, 15:25) <i>14:09 ± 02:22</i> N=10; d=55
Median (25th percentile, 75th percentile), <i>circular or arithmetic mean ± standard deviation</i> , N=participants; d=participant-days with a finite value for that metric.							
¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.							
² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.							
³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.							

Table continued, column panel A

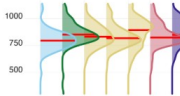
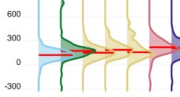
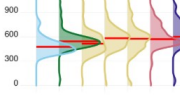
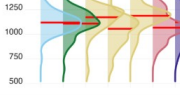
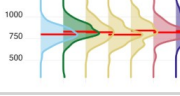
Metric descriptive summary (near eye)								
Metric ¹	Unit	Overall	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Duration								
Time above 1,000 lx melEDI Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	HH:MM	00:41 (00:11, 01:33) 01:10 ± 01:28 N=141; d=816	00:27 (00:06, 00:53) 00:38 ± 00:47 N=23; d=129	00:30 (00:06, 01:04) 00:44 ± 00:51 N=17; d=101	00:19 (00:09, 00:43) 00:32 ± 00:33 N=6; d=32	00:43 (00:17, 01:14) 00:58 ± 00:57 N=15; d=81	Symlog (base 10; threshold 1)	
Time above 250 lx melEDI during wake Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	HH:MM	02:25 (00:51, 04:37) 03:00 ± 02:29 N=141; d=737	02:43 (00:54, 04:43) 03:02 ± 02:19 N=23; d=113	01:30 (00:41, 03:26) 02:09 ± 01:57 N=17; d=94	01:39 (00:33, 02:40) 01:48 ± 01:21 N=6; d=30	01:14 (00:25, 02:37) 01:45 ± 01:46 N=15; d=78	Symlog (base 10; threshold 1)	
Time below 10 lx melEDI before sleep Low-light time before bed; limits evening melatonin suppression and circadian delay.	HH:MM	01:53 (01:01, 02:35) 01:51 ± 01:02 N=139; d=655	01:43 (00:58, 02:50) 01:50 ± 01:05 N=23; d=105	01:31 (00:49, 02:15) 01:38 ± 01:00 N=17; d=82	02:16 (01:15, 02:39) 01:56 ± 01:00 N=6; d=25	02:30 (01:28, 02:52) 02:09 ± 00:57 N=14; d=58	Symlog (base 10; threshold 1)	
Time below 1 lx melEDI during sleep Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	HH:MM	07:08 (05:58, 08:15) 07:04 ± 01:57 N=141; d=778	08:06 (07:10, 09:10) 08:02 ± 01:57 N=23; d=118	06:45 (05:08, 08:15) 06:40 ± 02:07 N=17; d=99	06:56 (06:06, 07:36) 06:39 ± 01:43 N=6; d=31	07:31 (06:31, 09:00) 07:35 ± 02:15 N=15; d=81	Symlog (base 10; threshold 1)	
Longest period above 250 lx melEDI Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	HH:MM	00:38 (00:17, 01:12) 00:55 ± 01:00 N=141; d=816	00:38 (00:16, 01:04) 00:45 ± 00:36 N=23; d=129	00:34 (00:14, 00:55) 00:43 ± 00:41 N=17; d=101	00:28 (00:16, 00:43) 00:30 ± 00:20 N=6; d=32	00:23 (00:15, 00:49) 00:38 ± 00:45 N=15; d=81	Symlog (base 10; threshold 1)	
Dynamics								
Interdaily stability Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	—	0.308 (0.248, 0.38) 0.318 ± 0.095 N=141; d=816	0.352 (0.321, 0.455) 0.368 ± 0.092 N=23; d=129	0.278 (0.254, 0.394) 0.323 ± 0.104 N=17; d=101	0.342 (0.315, 0.473) 0.37 ± 0.118 N=6; d=32	0.293 (0.255, 0.354) 0.299 ± 0.062 N=15; d=81	Identity	
Intradaily variability Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	—	1.253 (0.93, 1.502) 1.229 ± 0.389 N=141; d=816	1.251 (0.999, 1.497) 1.251 ± 0.413 N=23; d=129	1.359 (0.95, 1.546) 1.307 ± 0.415 N=17; d=101	1.399 (1.282, 1.483) 1.355 ± 0.256 N=6; d=32	1.324 (1.029, 1.595) 1.344 ± 0.413 N=15; d=81	Identity	
Exposure history								
melEDI dose Intensity–duration–weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	klx·h	4.96 (1.936, 12.313) 10.358 ± 16.216 N=141; d=761	3.317 (1.703, 6.29) 6.115 ± 8.79 N=23; d=125	4.35 (1.791, 10.625) 7.49 ± 8.898 N=17; d=96	2.195 (1.339, 4.117) 3.281 ± 2.573 N=6; d=30	6.045 (2.49, 14.907) 10.016 ± 10.799 N=15; d=76	Symlog (base 10; threshold 1)	

Table continued, column panel A

Metric descriptive summary (near eye)								
Metric ¹	Unit	Overall	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Level								
Mean melEDI Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	lx	5.154 (2.831, 9.225) <i>7.587 ± 9.496</i> N=141; d=816	3.505 (1.469, 6.211) <i>4.166 ± 3.272</i> N=23; d=129	5.857 (3.726, 7.986) <i>7.778 ± 6.946</i> N=17; d=101	4.325 (3.585, 6.77) <i>5.248 ± 2.85</i> N=6; d=32	2.144 (0.744, 4.042) <i>3.019 ± 2.896</i> N=15; d=81	Symlog (base 10; threshold 1)	
Brightest 10 h mean Mean of the brightest 10 hours; reflects the strength of the main daytime light episode.	lx	110.566 (41.513, 243.212) <i>214.866 ± 508.56</i> N=141; d=816	101.102 (24.355, 218.457) <i>136.312 ± 134.07</i> N=23; d=129	83.24 (53.17, 151.786) <i>120.172 ± 101.545</i> N=17; d=101	68.143 (45.331, 132.458) <i>96.735 ± 72.311</i> N=6; d=32	46.66 (9.242, 103.175) <i>82.489 ± 115.07</i> N=15; d=81	Symlog (base 10; threshold 1)	
Darkest 10 h mean Mean of the darkest 10 hours; lower values during the biological night favour melatonin preservation and sleep.	lx	0.103 (0.02, 0.253) <i>0.243 ± 0.513</i> N=141; d=816	0.022 (0, 0.08) <i>0.054 ± 0.075</i> N=23; d=129	0.152 (0.052, 0.481) <i>0.475 ± 0.882</i> N=17; d=101	0.133 (0.055, 0.309) <i>0.411 ± 1.044</i> N=6; d=32	0.01 (0, 0.092) <i>0.059 ± 0.104</i> N=15; d=81	Symlog (base 10; threshold 1)	
Spectrum								
Melanopic daylight efficacy ratio Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	—	0.724 (0.643, 0.795) <i>0.724 ± 0.116</i> N=137; d=687	0.668 (0.629, 0.721) <i>0.673 ± 0.07</i> N=22; d=83	0.747 (0.634, 0.834) <i>0.735 ± 0.122</i> N=17; d=91	0.719 (0.661, 0.761) <i>0.706 ± 0.067</i> N=6; d=30	0.755 (0.729, 0.82) <i>0.758 ± 0.097</i> N=12; d=48	Identity	

1150

Table continued, column panel A

Metric descriptive summary (near eye)								
Metric ¹	Unit	Overall	Madrid (ES)	Izmir (TR)	San José (CR)	Kumasi (GH)	Scaling ²	Distribution ³
Timing								
Midpoint of the brightest 10 hours Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	HH:MM clock time	13:44 (12:48, 15:00) 13:56 ± 01:51 N=141; d=816	13:44 (13:06, 15:20) 14:10 ± 01:51 N=23; d=129	14:10 (13:32, 15:21) 14:25 ± 01:37 N=17; d=101	12:34 (11:49, 14:02) 12:55 ± 01:44 N=6; d=32	12:49 (12:06, 13:39) 13:03 ± 01:58 N=15; d=81	Circular clock	
Midpoint of the darkest 10 hours Centre time of the darkest 10 hours; indexes the main daily darkness cue.	HH:MM clock time	02:54 (02:01, 03:50) 02:58 ± 01:41 N=141; d=816	03:39 (02:43, 04:34) 03:34 ± 01:49 N=23; d=129	03:31 (02:33, 04:14) 03:26 ± 01:25 N=17; d=101	01:45 (01:05, 03:01) 02:10 ± 01:27 N=6; d=32	02:33 (02:04, 03:17) 02:36 ± 01:36 N=15; d=81	Circular clock	
First light timing above 250 lx melEDI First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	HH:MM clock time	09:08 (08:02, 10:40) 09:26 ± 02:22 N=140; d=727	09:32 (08:40, 11:21) 10:05 ± 02:18 N=23; d=118	10:01 (09:06, 11:52) 10:26 ± 02:28 N=17; d=95	08:11 (07:26, 09:57) 08:36 ± 01:30 N=6; d=29	08:43 (08:01, 10:10) 09:19 ± 01:52 N=15; d=61	Circular clock	
Last light timing above 250 lx melEDI Last bright-light exposure; later timing may delay circadian phase and sleep onset.	HH:MM clock time	18:08 (16:27, 19:42) 18:03 ± 02:36 N=141; d=687	17:48 (16:46, 19:19) 18:01 ± 02:21 N=23; d=106	18:45 (17:02, 19:38) 18:23 ± 02:15 N=17; d=89	15:45 (13:20, 16:38) 15:22 ± 02:38 N=6; d=28	15:54 (14:34, 17:25) 15:46 ± 02:13 N=15; d=66	Circular clock	
Mean timing of exposure above 250 lx melEDI Average bright-light timing; summarizes the phase of daily circadian stimulation.	HH:MM clock time	13:29 (12:29, 14:39) 13:31 ± 01:52 N=141; d=742	13:43 (12:55, 14:29) 13:47 ± 01:33 N=23; d=117	13:58 (12:54, 14:58) 14:04 ± 01:39 N=17; d=94	11:30 (10:28, 12:38) 11:26 ± 01:36 N=6; d=31	12:17 (11:04, 13:04) 12:11 ± 01:36 N=15; d=71	Circular clock	
Median (25th percentile, 75th percentile), <i>circular or arithmetic mean ± standard deviation</i> , N=participants; d=participant-days with a finite value for that metric.								
¹ The brief meaning and relevance notes describe established physiological constructs; this descriptive table does not estimate individual health effects.								
² Scaling describes the distribution column only; printed values remain on their stated scale. Symlog uses base 10 with a linear region through 1 in the stated unit; Identity is linear; Circular clock unwraps values around the metric-specific clock centre.								
³ Red lines indicate site medians. Symlog and circular-clock rows are transformed only for plotting.								

1151

1152 **Supplementary Table S2.** Definitions, exact support, median (interquartile range) summaries, and density displays for the 17 near-eye personal light-exposure

1153 metrics overall and by site. Values use three decimal places where applicable. Metrics are distinct summaries and should not be interpreted as interchangeable

1154 outcomes.

Site ¹	Minutes in the recommended range: ²				Share of all eligible real minutes: ³			
	Daytime	Pre-sleep	Sleep ⁴	Total ⁵	Daytime	Pre-sleep	Sleep	Unclassified ⁶
Overall	24.0% 137,792 / 573,712	63.3% 81,894 / 129,390	87.7% 336,052 / 383,366	51.2% 555,738 / 1,086,468	51.6% 606,215 / 1,175,160	11.9% 140,152 / 1,175,160	32.6% 383,607 / 1,175,160	3.8% 45,186 / 1,175,160
Borås (SE)	33.2% 19,225 / 57,952	69.9% 8,914 / 12,759	93.4% 33,152 / 35,489	57.7% 61,291 / 106,200	54.0% 60,642 / 112,260	12.2% 13,704 / 112,260	31.7% 35,549 / 112,260	2.1% 2,365 / 112,260
Delft (NL)	29.6% 14,760 / 49,852	66.7% 7,181 / 10,774	86.4% 29,915 / 34,621	54.4% 51,856 / 95,247	47.0% 52,755 / 112,260	10.9% 12,283 / 112,260	30.9% 34,682 / 112,260	11.2% 12,540 / 112,260
Dortmund (DE)	32.4% 22,631 / 69,900	59.6% 9,749 / 16,361	87.5% 42,829 / 48,928	55.6% 75,209 / 135,189	48.1% 74,094 / 154,080	11.3% 17,409 / 154,080	31.8% 48,928 / 154,080	8.9% 13,649 / 154,080
Tübingen (DE)	22.5% 25,082 / 111,396	61.8% 15,623 / 25,283	87.1% 61,030 / 70,029	49.2% 101,735 / 206,708	55.1% 119,150 / 216,120	12.4% 26,881 / 216,120	32.4% 70,089 / 216,120	0.0% 0 / 216,120
Munich (DE)	25.0% 10,681 / 42,685	60.6% 5,963 / 9,833	71.7% 21,485 / 29,951	46.2% 38,129 / 82,469	53.0% 45,831 / 86,400	12.3% 10,618 / 86,400	34.7% 29,951 / 86,400	0.0% 0 / 86,400
Madrid (ES)	24.2% 21,067 / 87,030	62.7% 12,766 / 20,366	95.9% 58,374 / 60,891	54.8% 92,207 / 168,287	48.4% 89,948 / 185,880	11.7% 21,684 / 185,880	32.8% 60,891 / 185,880	7.2% 13,357 / 185,880
Izmir (TR)	17.8% 12,708 / 71,562	55.1% 9,221 / 16,736	79.0% 39,936 / 50,556	44.6% 61,865 / 138,854	51.5% 74,926 / 145,440	12.3% 17,883 / 145,440	34.8% 50,556 / 145,440	1.4% 2,075 / 145,440
San José (CR)	13.8% 3,266 / 23,640	64.6% 3,124 / 4,839	87.1% 12,438 / 14,275	44.0% 18,828 / 42,754	54.9% 25,285 / 46,080	11.5% 5,320 / 46,080	31.0% 14,275 / 46,080	2.6% 1,200 / 46,080
Kumasi (GH)	14.0% 8,372 / 59,695	75.2% 9,353 / 12,439	95.5% 36,893 / 38,626	49.3% 54,618 / 110,760	54.5% 63,584 / 116,640	12.3% 14,370 / 116,640	33.2% 38,686 / 116,640	0.0% 0 / 116,640

¹ Near eye participant and participant-day sample sizes are reported in the participant table; every percentage cell gives its exact minute denominator.

² Recommended ranges follow Brown et al. (2022): Daytime ≥250 lx melanopic EDI, Pre-sleep ≤10 lx melanopic EDI, and Sleep ≤1 lx melanopic EDI. Daytime, Pre-sleep, and Sleep identify the recommendation windows. Each grey numerator/denominator gives minutes within the applicable recommendation range over all valid one-minute observations in that window. Sleep describes the bedside sleep environment.

³ Each grey relevant / all count gives minutes in the displayed diary state over all eligible real minutes before classification by diary state or measurement availability.

⁴ During diary-defined sleep the bedside sensor describes the sleep environment rather than direct ocular exposure.

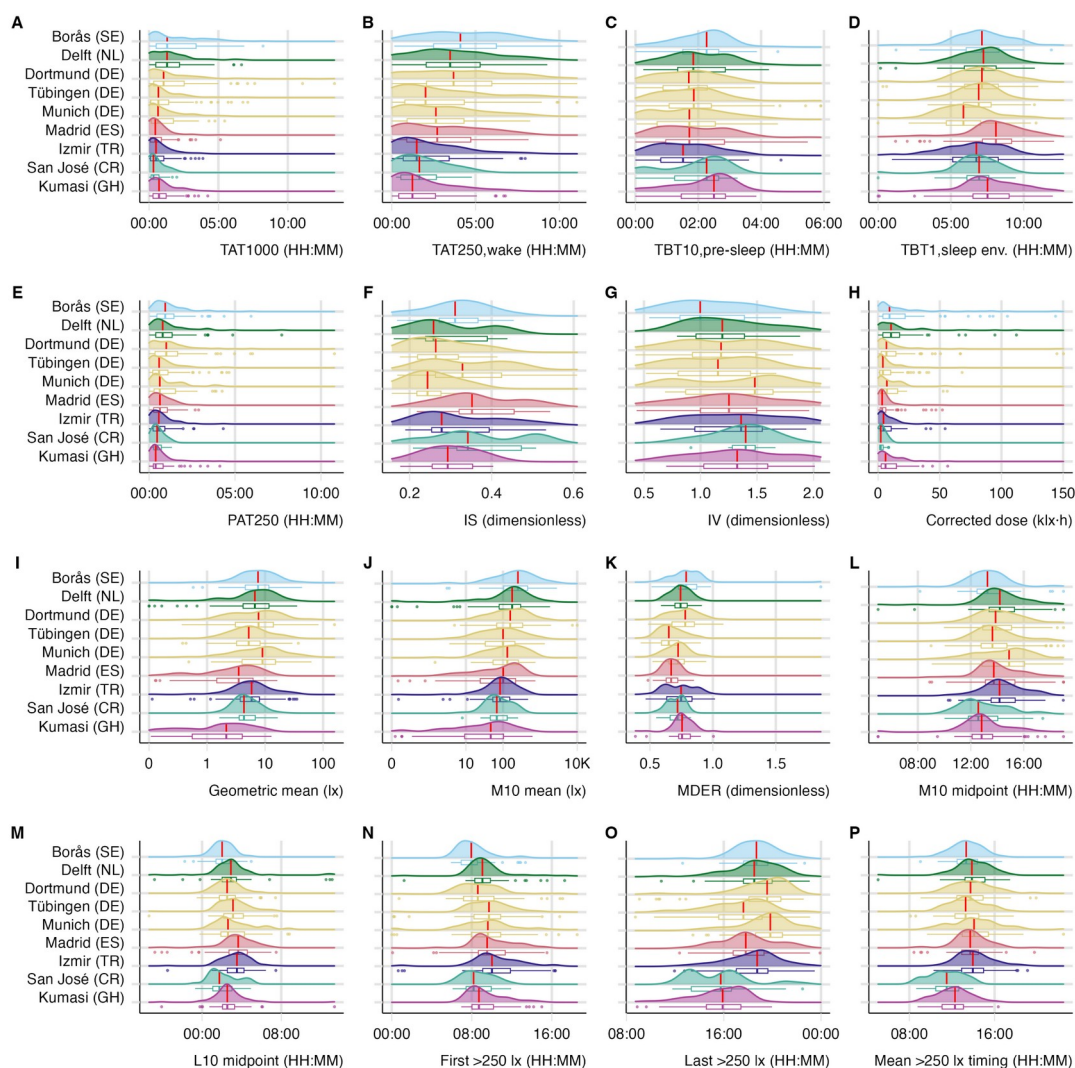
⁵ Total pools all classified valid minutes.

⁶ Unclassified time can result from missing diary state, missing or removed melEDI measurements, or both.

1157 **Supplementary Table S3.** Fractions of valid near-eye minutes meeting each Brown et al. recommendation range, with explicit numerators and denominators.
1158 Daytime, Pre-sleep, and Sleep identify the recommendation windows. The Sleep summaries use near-eye illuminance recorded during sleep and describe the
1159 bedside sleep environment rather than verified ocular exposure. These pooled-minute summaries are descriptive and differ from the modelled
1160 recommendation-window-period estimates in main Table 2.

1161 **Supplementary Figure S1. Near-eye metric distributions**

1162

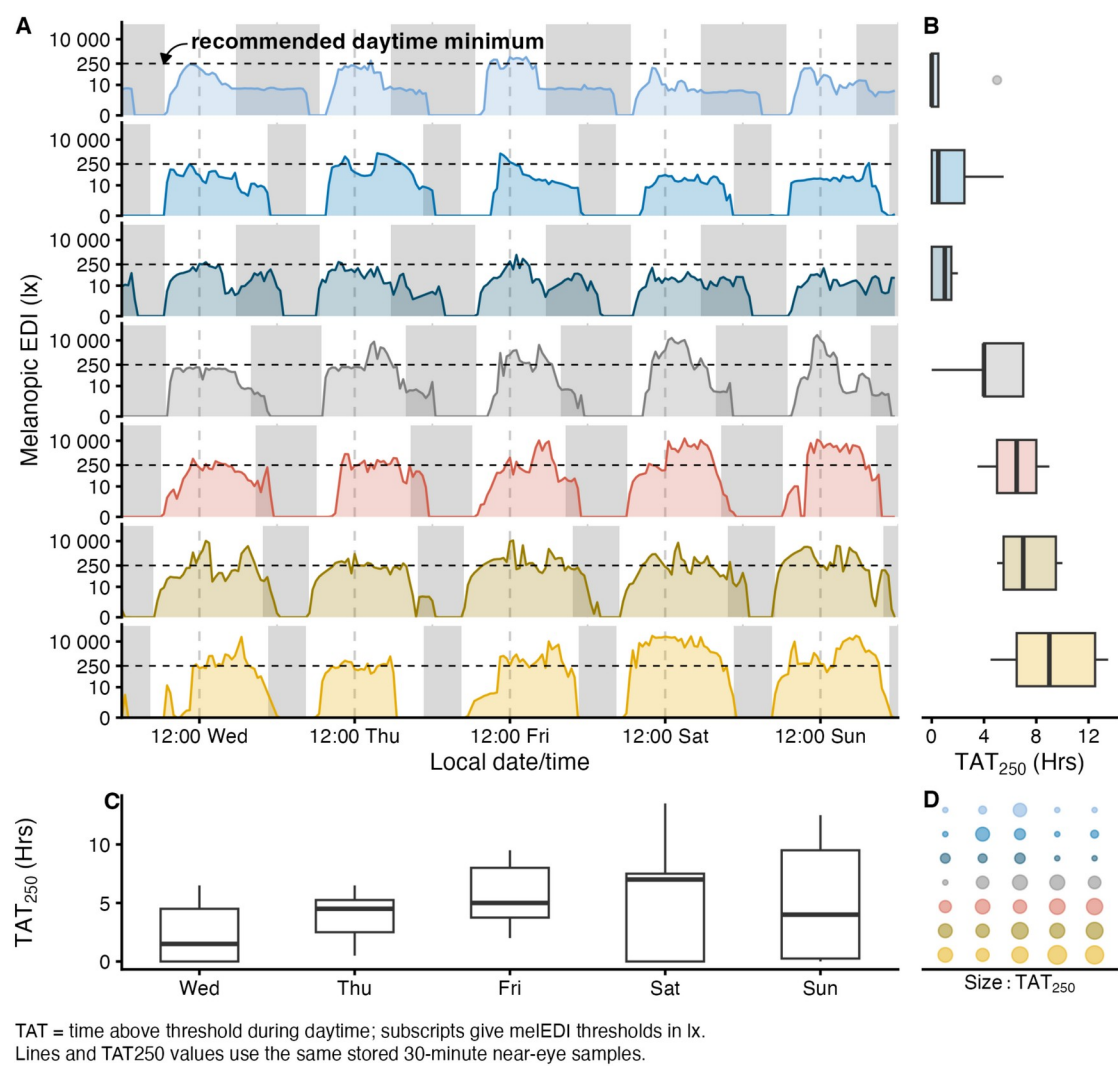


1163

1164 **Supplementary Figure S1. Detailed near-eye metric distributions by country-coded study**
1165 **site.**

1166 **Supplementary Figure S2. Worked metric derivation**

1167



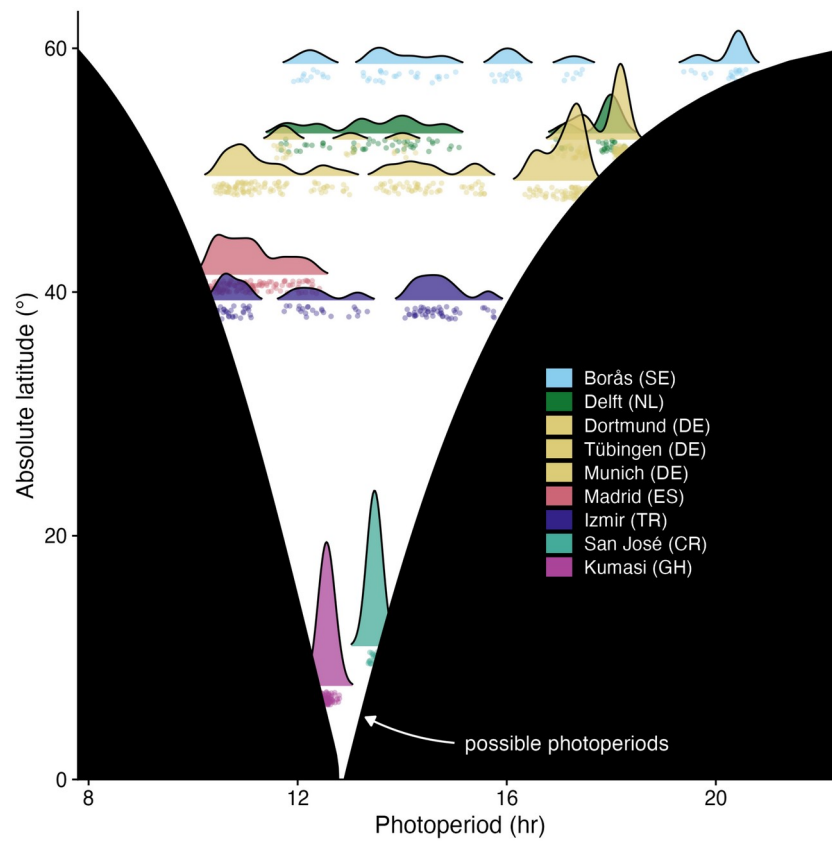
1168

1169 Supplementary Figure S2. Examples linking 30-minute time series to the daily time-

1170 above-250-lx metric.

1171 **Supplementary Figure S3. Latitude and civil-photoperiod**
1172 **measurement range**

1173



1174

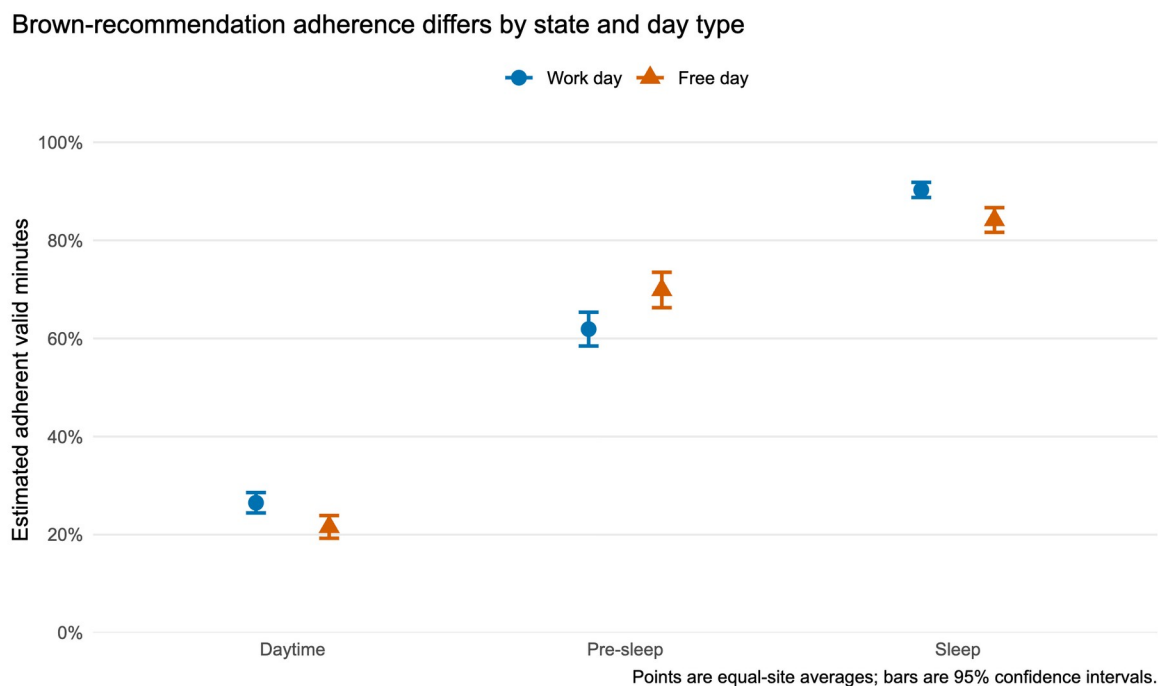
1175 Supplementary Figure S3. Observed civil photoperiod and theoretical bounds by absolute
1176 latitude.

1177 Recommendation adherence

1178 Daytime, Pre-sleep, and Sleep identify the Brown et al. recommendation windows.

1179 Supplementary Figure S4. Site-average recommendation adherence

1180

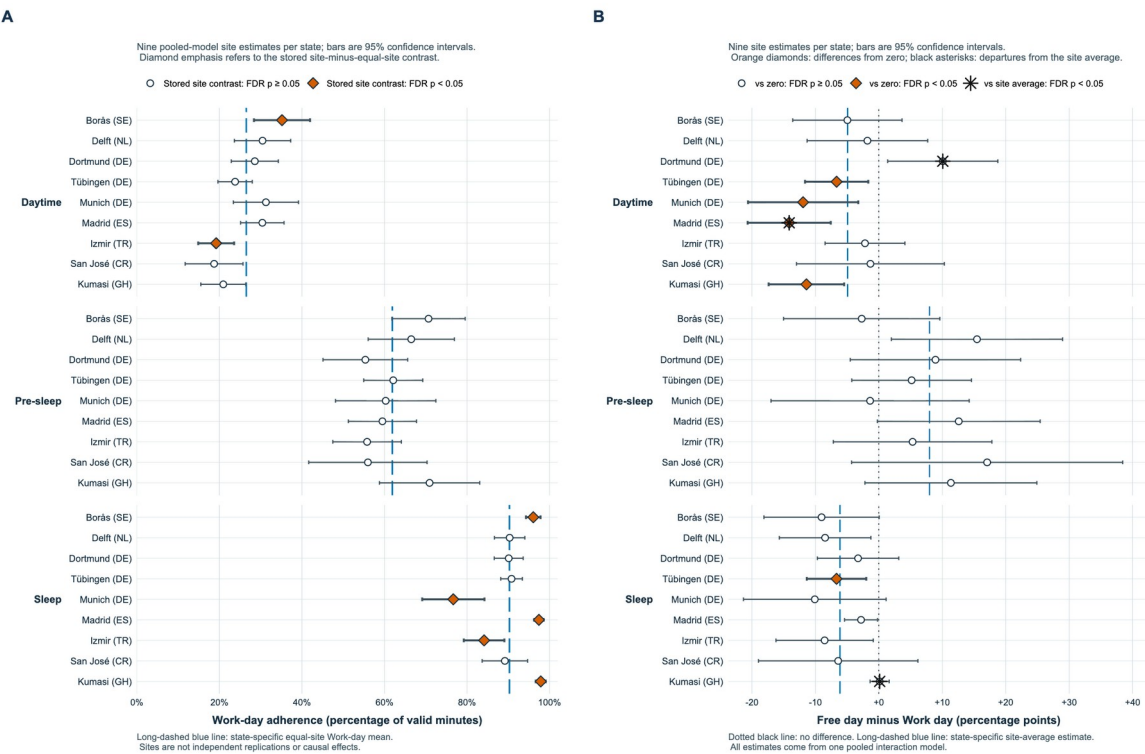


1181

1182 Supplementary Figure S4. Recommendation adherence in the Daytime, Pre-sleep and Sleep
1183 recommendation windows on Work and Free days. Points are equal-site average
1184 percentages of valid minutes meeting the Brown et al. recommendation; bars are 95%
1185 confidence intervals. Daytime, Pre-sleep and Sleep identify the three Brown et
1186 al. recommendation windows. The Sleep estimate uses near-eye illuminance recorded
1187 during sleep and describes the bedside sleep environment rather than verified ocular
1188 exposure.

1189 **Supplementary Figure S5. Site-specific adherence and day-type**
1190 **differences**

1191



1192

1193 Supplementary Figure S5. Site-specific recommendation adherence across Brown et
1194 al. recommendation windows. A, Work-day adherence, with diamonds identifying deviations
1195 from the window-specific site-average Work-day estimate that met its false-discovery-rate
1196 (FDR) criterion. B, Free-minus-Work differences, with diamonds identifying differences from
1197 zero and asterisks identifying departures from the window-specific site-average difference in
1198 a separate 27-comparison FDR family. Daytime, Pre-sleep and Sleep identify the Brown et
1199 al. recommendation windows. Points are estimates and bars are 95% confidence intervals.
1200 Sites are components of one pooled model, not independent replications or causal effects of
1201 location. The Sleep-window estimates use near-eye illuminance recorded during sleep and
1202 describe the bedside sleep environment rather than verified ocular exposure.

1203 **Supplementary Table S4. Exploratory cross-window recommendation-adherence associations**

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Within participant, linked cycle				
Sleep	-0.26 (-0.95 to 0.43)	0.453	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.
Pre-sleep	-1.09 (-2.75 to 0.56)	0.258	Four prespecified primary cross-window tests	No retained evidence that a cycle with 10 percentage points higher Daytime adherence had different Pre-sleep adherence; the day-level claim is withheld because serial dependence remains unresolved.

1204

Table continued

Exploratory cross-window recommendation-adherence associations				
Outcome window	Difference per +10 percentage points Daytime adherence, percentage points (95% CI)	FDR-adjusted p	FDR-correction set/family	Interpretation for recommendation adherence
Between participants, observed monitoring-period average				
Sleep	-2.52 (-3.55 to -1.48)	<0.001	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 2.52 percentage points lower average Sleep adherence over the monitoring period.
Pre-sleep	-3.59 (-5.94 to -1.25)	0.005	Four prespecified primary cross-window tests	Participants with 10 percentage points higher average Daytime adherence had 3.59 percentage points lower average Pre-sleep adherence over the monitoring period.
All four tests form one FDR-correction set. Site-standardized estimates give nine sites equal weight and use 50:50 Work/Free weighting. The primary sample contained 1,376 target-window rows from 140 participants and 761 linked cycles. Within-participant claims are withheld because temporal dependence remains unresolved. Between-participant estimates describe observed monitoring-period averages, not rankings, stable traits, or causal effects. Confidence intervals are not multiplicity adjusted.				

1205

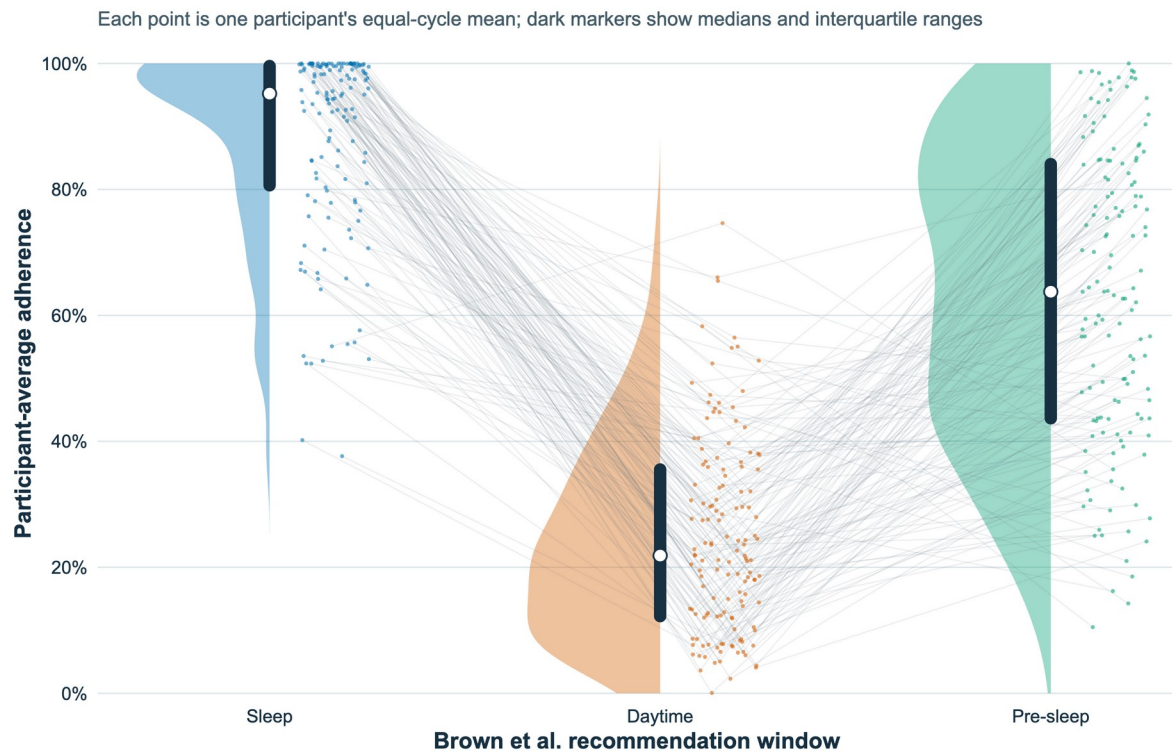
1206 **Supplementary Table S4.** Exploratory within-participant and between-participant associations across Brown et al. recommendation windows. Results are

1207 percentage-point differences per 10 percentage points higher Daytime adherence. The four tests form one FDR family. The within-participant day-level claim is

1208 withheld because serial dependence remains unresolved. Between-participant results describe observed monitoring-period averages and do not establish
1209 rankings, stable traits, or causal effects.

1210 **Supplementary Figure S6. Anonymous participant**
 1211 **recommendation-adherence profiles**

1212



1213

1214 Supplementary Figure S6. Participant-average recommendation adherence across the
 1215 Brown et al. recommendation windows. Daytime, Pre-sleep, and Sleep are the window
 1216 labels. Each colored point is one anonymous participant's equal-cycle mean within that
 1217 window. Faint lines connect the same anonymous participant across the three window
 1218 summaries; they are not time trajectories or ranks. Half-violin shapes show the window
 1219 distributions, and dark bars with white points show interquartile ranges and medians.
 1220 Because each window has its own Brown et al. threshold and direction, absolute
 1221 percentages should be interpreted within window.

1222 **Daily architecture**

1223 **Supplementary Table S5. Near-eye fitted-curve dispersion and full-model R² allocation**

Near-eye fitted-curve dispersion and conditional Shapley allocation			
	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.611 (0.550 to 0.657)	78.9% (73.6% to 82.6%)
Site pattern	0.100 (0.041 to 0.139)	0.016 (0.007 to 0.026)	2.0% (0.9% to 3.5%)
Participant pattern	0.180 (0.126 to 0.214)	0.100 (0.079 to 0.125)	12.9% (10.1% to 16.4%)
Participant-day shift	0.019 (0.010 to 0.024)	0.048 (0.035 to 0.064)	6.2% (4.4% to 8.6%)
Participant pattern + day shift	0.200 (0.140 to 0.230)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.797 (1.157 to 4.332)	6.41 (3.85 to 15.14)	Not applicable
(Participant + day) / site	1.991 (1.291 to 4.767)	9.49 (5.77 to 21.81)	90.5% (85.2% to 95.6%)

Full-model in-sample R² was 0.775 (0.739 to 0.804). Fitted-curve variation is in squared log₁₀(melEDI + 0.1 lx) prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R² with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

1224

1225 **Supplementary Table S5.** Near-eye fitted-curve dispersion and Shapley allocation of full-model in-sample R². Fitted-curve variation is in squared

1226 log₁₀(melanopic EDI + 0.1 lx) prediction units; dispersion and R²-share ratios are unitless. The allocation includes the shared local-clock curve in every

1227 component model as the baseline. Intervals are 95% hierarchical cluster-bootstrap percentile intervals from 2,000 replicates and are conditional on the fitted
 1228 models. Dispersion and R² allocation describe different estimands and should not be added or interpreted causally.

1229 **Supplementary Table S6. Chest fitted-curve dispersion and full-model R² allocation**

Complementary chest fitted-curve dispersion and conditional Shapley allocation

	Fitted-curve result (95% CI)	Conditional Shapley result (95% CI)	Share (95% CI)
Component summaries			
Shared local-clock curve	Not applicable	0.583 (0.523 to 0.629)	79.1% (73.6% to 82.8%)
Site pattern	0.101 (0.049 to 0.116)	0.015 (0.008 to 0.024)	2.1% (1.1% to 3.3%)
Participant pattern	0.148 (0.100 to 0.199)	0.087 (0.071 to 0.110)	11.9% (9.5% to 15.3%)
Participant-day shift	0.035 (0.020 to 0.039)	0.051 (0.040 to 0.067)	7.0% (5.3% to 9.4%)
Participant pattern + day shift	0.183 (0.127 to 0.228)	Not separately allocated	Not applicable
Participant-to-site comparisons			
Participant / site	1.465 (0.973 to 3.179)	5.69 (3.72 to 10.68)	Not applicable
(Participant + day) / site	1.809 (1.219 to 3.736)	9.02 (6.01 to 17.03)	90.0% (85.7% to 94.5%)

Full-model in-sample R² was 0.737 (0.703 to 0.766). Fitted-curve variation is in squared log₁₀(meEDI + 0.1 lx) prediction units; curve and Shapley ratios are unitless. Conditional Shapley values allocate in-sample R² with the shared local-clock curve as the mandatory baseline. Intervals are 95% percentile hierarchical cluster-bootstrap intervals from 2,000 replicates. Dispersion and Shapley credit are different estimands and must not be added or interpreted causally.

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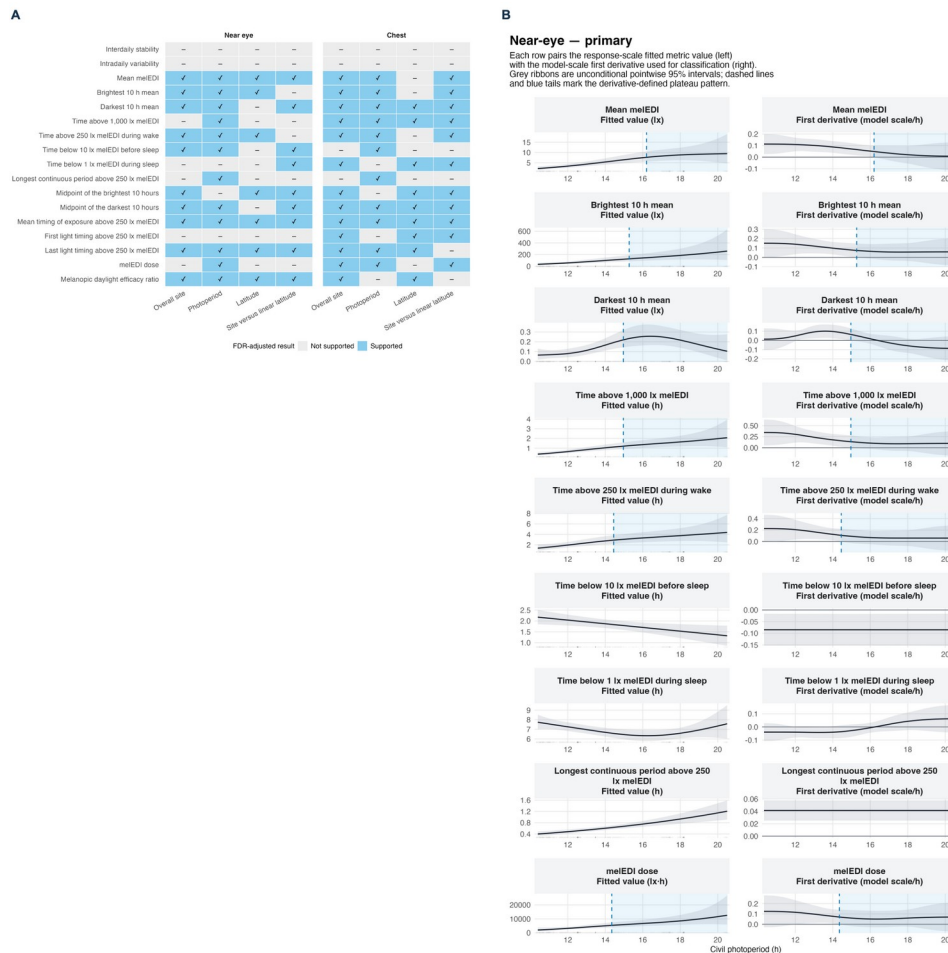
1231 **Supplementary Table S6.** Complementary chest fitted-curve dispersion and Shapley allocation of full-model in-sample R². The quantities, bootstrap intervals
 1232 and interpretation follow Supplementary Table S5; chest and near-eye measurements remain distinct sensor-position estimands.

1233 Geographic and civil-photoperiod context

1234 Supplementary Figure S7. Geographic support and nonlinear

1235 photoperiod classifications

1236



1237

1238 Supplementary Figure S7. Geographic and civil-photoperiod associations across near-eye

1239 personal light-exposure metrics. A, Evidence retained after FDR adjustment for overall site,

1240 civil photoperiod, latitude, and site-versus-linear-latitude adequacy. B, Nonlinear civil-

1241 photoperiod associations and their fitted slopes. A qualifying transition denotes an increase

1242 followed by a sustained near-flat tail under the specified slope rule. These descriptive

1243 associations do not identify a mechanism, and no physiological or environmental ceiling was

1244 identified.

1245 **Supplementary Table S7. Geographic and photoperiod associations**

		Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Overall site FDR-adjusted p	Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)		
Dynamics							
Interdaily stability	0.228	×0.98 [0.94–1.02]	0.401	×1.01 [0.96–1.06]	0.903	0.175	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Intradaily variability	0.483	-0.02 [-0.06–0.02]	0.327	-0.04 [-0.08–0.01]	0.139	0.723	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Level							
Mean melEDI	<0.001	×1.18 [1.11–1.26]	<0.001	×1.15 [1.07–1.25]	0.002	0.032	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Brightest 10 h mean	0.011	×1.24 [1.13–1.36]	<0.001	×1.24 [1.11–1.39]	0.001	0.325	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Darkest 10 h mean	<0.001	×1.09 [1.03–1.14]	0.002	×1.04 [0.97–1.11]	0.400	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

Table continued

	Overall site FDR-adjusted p	Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Duration							
Time above 1,000 lx melEDI	0.071	×1.21 [1.13–1.29]	<0.001	×1.03 [0.94–1.12]	0.769	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Time above 250 lx melEDI during wake	0.007	×1.14 [1.07–1.20]	<0.001	×1.12 [1.04–1.20]	0.013	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 737$
Time below 10 lx melEDI before sleep	0.049	-0.13 [-0.20--0.07] h	<0.001	0.00 [-0.09–0.09] h	0.962	0.040	$n_{\text{participants}} = 139$; $n_{\text{participant-days}} = 655$
Time below 1 lx melEDI during sleep	0.058	×0.99 [0.97–1.01]	0.227	×1.00 [0.97–1.02]	0.903	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 778$
Longest continuous period above 250 lx melEDI	0.398	×1.12 [1.07–1.19]	<0.001	×1.06 [0.99–1.13]	0.139	0.626	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

Table continued

	Overall site FDR-adjusted p	Photoperiod		Latitude		Site-versus-latitude FDR-adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR-adjusted p	Latitude association per 10° (95% CI)	Latitude FDR-adjusted p		
Timing							
Midpoint of the brightest 10 hours	<0.001	0.05 [-0.05–0.15] h	0.305	0.19 [0.06–0.32] h	0.013	0.010	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 816
Midpoint of the darkest 10 hours	0.020	-0.15 [-0.25–-0.05] h	0.004	0.12 [-0.00–0.24] h	0.114	0.043	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 816
Mean timing of exposure above 250 lx melEDI	<0.001	0.12 [0.03–0.21] h	0.013	0.33 [0.21–0.46] h	<0.001	<0.001	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 742
First light timing above 250 lx melEDI	0.112	-0.10 [-0.25–0.05] h	0.227	0.02 [-0.17–0.21] h	0.903	0.077	<i>n</i> _{participants} = 140; <i>n</i> _{participant-days} = 727
Last light timing above 250 lx melEDI	<0.001	0.32 [0.19–0.46] h	<0.001	0.46 [0.27–0.64] h	<0.001	<0.001	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 687
Exposure history							
melEDI dose	0.209	×1.28 [1.17–1.39]	<0.001	×1.02 [0.91–1.13]	0.903	0.156	<i>n</i> _{participants} = 141; <i>n</i> _{participant-days} = 761
Spectrum							
Melanopic daylight efficacy ratio	<0.001	0.02 [0.02–0.03]	<0.001	-0.01 [-0.02–-0.00]	0.013	0.010	<i>n</i> _{participants} = 137; <i>n</i> _{participant-days} = 702
This summary combines the model-level decisions, photoperiod and latitude associations, 95% CIs, and exact fitted samples. Each displayed p-value is FDR-adjusted within its explicitly labelled complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Full raw p-values and test statistics remain in the detailed tables below. Exact fitted samples use italic n with participant and participant-day subscripts. For the 15 participant-day models, the participant-day count equals the number of fitted observations. The two participant-level dynamics models use one fitted observation per participant and retain the participant-day count as contributing repeated-day support. All models include nine sites.							

1249 **Supplementary Table S7.** Overall-site, civil-photoperiod and absolute-latitude associations for the 17 near-eye personal light-exposure metrics. Each displayed
1250 p value is FDR-adjusted within its stated 17-test family. Site and latitude were evaluated in separate models because absolute latitude is fixed within and
1251 confounded with study site.

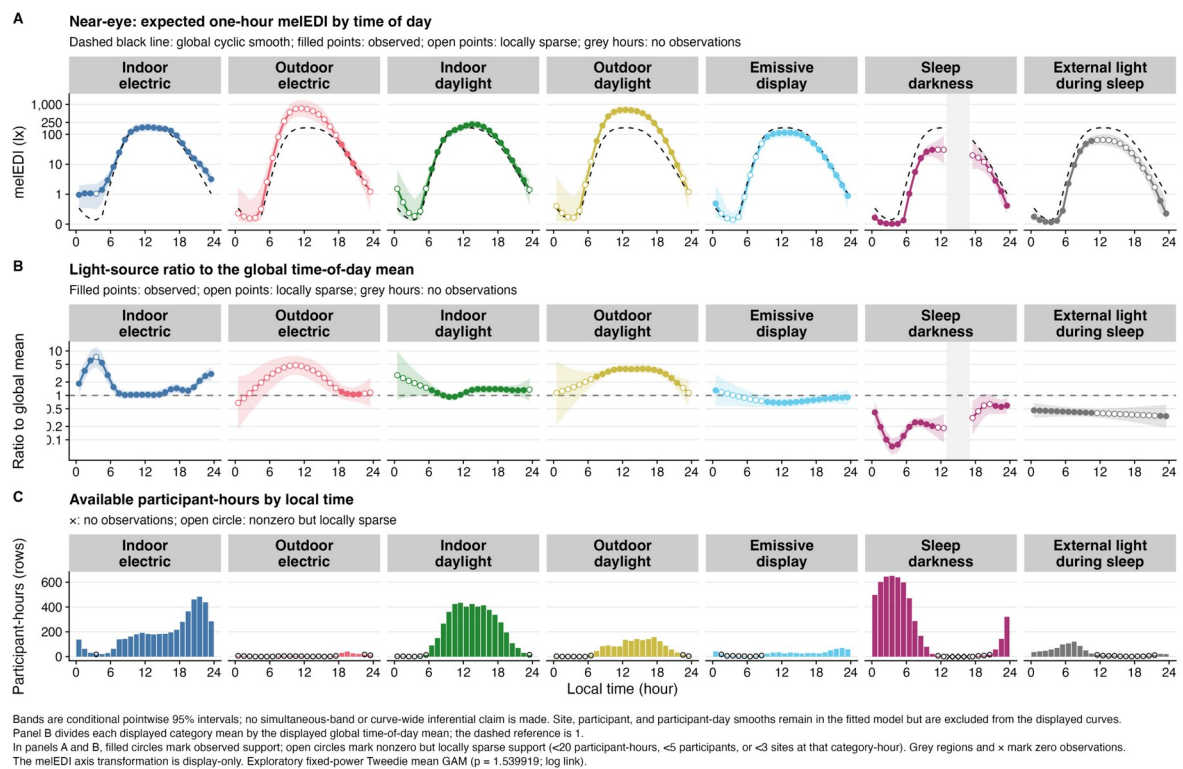
	Qualifying transition	Qualifying transition bracket (h)	Endpoint derivative	Interpretation
Light level				
Mean melEDI	Yes	16.10–16.20	0.008 (95% CI -0.111 to 0.127)	Detected increase followed by a sustained zero-compatible tail
Brightest 10 h mean	Yes	15.17–15.27	0.059 (95% CI -0.083 to 0.201)	Detected increase followed by a sustained zero-compatible tail
Darkest 10 h mean	Yes	14.86–14.96	-0.086 (95% CI -0.216 to 0.044)	Detected increase followed by a sustained zero-compatible tail
Duration and continuous period				
Time above 1,000 lx melEDI	Yes	14.86–14.96	0.098 (95% CI -0.176 to 0.371)	Detected increase followed by a sustained zero-compatible tail
Time above 250 lx melEDI during wake	Yes	14.35–14.45	0.060 (95% CI -0.159 to 0.279)	Detected increase followed by a sustained zero-compatible tail
Time below 10 lx melEDI before sleep	No	—	-0.085 (95% CI -0.154 to -0.016)	No preceding detected increase
Time below 1 lx melEDI during sleep	No	—	0.062 (95% CI -0.042 to 0.165)	No preceding detected increase
Longest continuous period above 250 lx melEDI	No	—	0.041 (95% CI 0.025 to 0.058)	Increase remained detected at the recorded maximum
Exposure history				
melEDI dose	Yes	14.24–14.35	0.070 (95% CI -0.069 to 0.209)	Detected increase followed by a sustained zero-compatible tail
A 'Yes' is the formal derivative-defined classification: the immediately preceding point has an interval wholly above zero, the next contains zero, and every later interval through the recorded maximum remains zero-compatible. Endpoint derivatives and 95% confidence intervals are on each model's linear-predictor scale, in model-scale units per hour of civil photoperiod.				

1254 **Supplementary Table S8.** Derivative-based classifications for nine near-eye personal light-exposure metrics across the observed civil-photoperiod range. A
1255 qualifying transition required the preceding fitted-slope interval to be wholly above zero, the next interval to include zero and every later interval through the
1256 recorded maximum to remain zero-compatible. These classifications are descriptive and do not identify a physiological or environmental ceiling.

1257 Immediate environments and routines

1258 Supplementary Figure S8. Light-source category and local-clock 1259 pattern of near-eye melanopic EDI

1260



1261

1262 **Supplementary Figure S8 (continued)**

D Near-eye light-source-by-site interaction model

Site-average estimates and site-specific deviation ratios; bold site cells pass the complete near-eye FDR adjustment

	Indoor electric P 138 · D 695 · H 4,629	Outdoor electric P 54 · D 118 · H 203	Indoor daylight P 138 · D 692 · H 4,804	Outdoor daylight P 135 · D 512 · H 1,557	Emissive display P 88 · D 235 · H 664	Sleep darkness P 130 · D 712 · H 5,225	External light during sleep P 66 · D 233 · H 853
Site-average estimate	85.1 lx 95% CI 71.6–101.1 1.000× reference	9.4 lx 95% CI 6.9–12.8 0.110× indoor 95% CI 0.078–0.156	198.8 lx 95% CI 169.2–233.4 2.335× indoor 95% CI 1.877–2.905	959.6 lx 95% CI 786.8–1170.2 11.274× indoor 95% CI 8.709–14.595	23.8 lx 95% CI 17.4–32.5 0.280× indoor 95% CI 0.197–0.398	2.2 lx 95% CI 1.5–3.3 0.026× indoor 95% CI 0.018–0.039	11.2 lx 95% CI 8.1–15.5 0.131× indoor 95% CI 0.091–0.189
● Borås (SE)	1.43× (1.02–1.99) FDR p 0.083	Not estimable	1.61× (1.12–2.32) FDR p 0.029	3.07× (1.92–4.90) FDR p <0.001	0.31× (0.16–0.57) FDR p 0.002	1.52× (0.39–5.90) FDR p 0.704	0.70× (0.21–2.36) FDR p 0.707
● Delft (NL)	1.83× (1.32–2.54) FDR p 0.002	Not estimable	1.46× (0.93–2.29) FDR p 0.190	1.58× (0.89–2.80) FDR p 0.214	7.17× (4.00–12.86) FDR p <0.001	0.94× (0.49–1.80) FDR p 0.894	Not estimable
● Dortmund (DE)	1.26× (0.69–2.29) FDR p 0.595	Not estimable	1.62× (0.83–3.17) FDR p 0.263	1.48× (0.91–2.40) FDR p 0.210	2.86× (1.66–4.93) FDR p 0.002	0.89× (0.45–1.75) FDR p 0.801	9.22× (2.50–34.03) FDR p 0.004
● Tübingen (DE)	0.56× (0.38–0.82) FDR p 0.009	20.42× (3.95–105.59) FDR p 0.002	1.04× (0.72–1.50) FDR p 0.866	0.89× (0.51–1.56) FDR p 0.776	0.25× (0.14–0.44) FDR p <0.001	0.78× (0.32–1.92) FDR p 0.710	1.59× (0.71–3.59) FDR p 0.406
● Munich (DE)	1.70× (0.63–4.57) FDR p 0.425	Not estimable	1.00× (0.65–1.55) FDR p 0.983	0.94× (0.55–1.61) FDR p 0.866	Not estimable	6.60× (2.11–20.69) FDR p 0.005	11.65× (7.19–18.85) FDR p <0.001
● Madrid (ES)	1.48× (1.15–1.90) FDR p 0.007	2.68× (1.56–4.61) FDR p 0.002	0.83× (0.57–1.19) FDR p 0.436	0.65× (0.33–1.26) FDR p 0.332	3.68× (2.24–6.03) FDR p <0.001	1.59× (0.30–8.32) FDR p 0.710	Not estimable
● Izmir (TR)	0.89× (0.52–0.92) FDR p 0.029	1.85× (0.84–4.07) FDR p 0.222	0.86× (0.65–1.13) FDR p 0.422	1.12× (0.65–1.94) FDR p 0.776	1.32× (0.70–2.47) FDR p 0.540	0.97× (0.43–2.18) FDR p 0.952	1.59× (0.92–2.76) FDR p 0.190
● San José (CR)	0.54× (0.38–0.77) FDR p 0.003	Not estimable	0.72× (0.49–1.05) FDR p 0.189	0.46× (0.21–1.01) FDR p 0.115	1.32× (0.65–2.66) FDR p 0.595	0.80× (0.25–1.42) FDR p 0.390	Not estimable
● Kumasi (GH)	0.58× (0.38–0.88) FDR p 0.029	6.24× (2.28–17.06) FDR p 0.002	0.49× (0.26–0.92) FDR p 0.065	0.50× (0.36–0.68) FDR p <0.001	1.27× (0.40–4.07) FDR p 0.776	0.17× (0.06–0.49) FDR p 0.004	Not estimable

Site-average estimates and comparisons use the light-source-by-site interaction model.
Site rows are deviation ratios around each category estimate; parentheses show 95% confidence intervals.
Bold site ratios and FDR p-values pass the complete 53-member near-eye family; unsupported cells are not estimable.
Near-eye frame: 17,935 participant-hours; 140 participants; 801 participant-days; nine sites.
Each accepted participant-hour has one primary light-source category; 1/k multi-label weighting does not apply.
Panels A–C exclude site, participant, and participant-day smooths from displayed curves; panel D uses stored interaction-model estimands.
P, participants; D, participant-days; H, participant-hours. Country-coded labels and row order provide non-colour site cues.

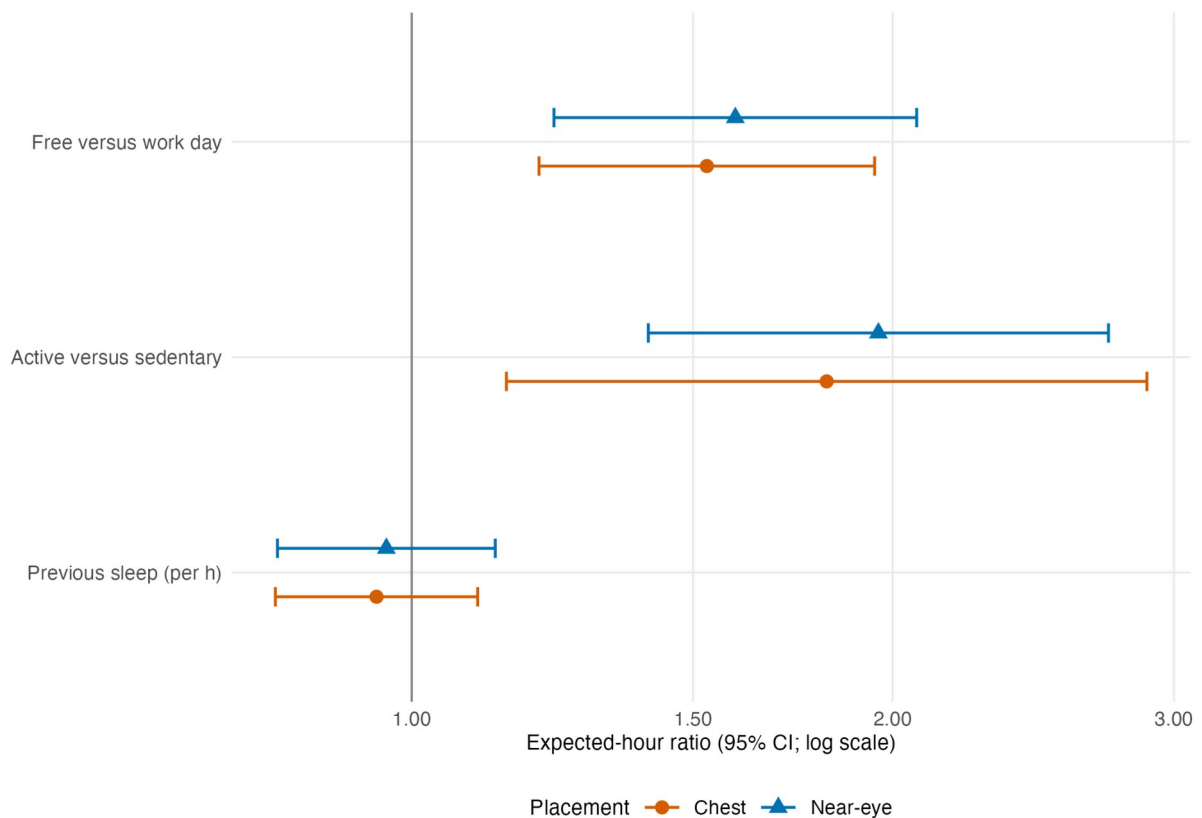
1263

1264 Supplementary Figure S8. Near-eye light exposure by reported light source across time of
1265 day and study site. A–C, Exploratory time-of-day analysis. Panel A shows expected one-
1266 hour melanopic EDI with clock-specific 95% confidence intervals; the dashed line is the site-
1267 average local-clock smooth. Panel B shows each light-source curve relative to that smooth,
1268 with 1 as the reference. Panel C shows available participant-hours; open circles indicate
1269 locally sparse support and grey gaps indicate no observations. D, Site-average category
1270 estimates and site-specific deviation ratios from the light-source-by-site interaction model,
1271 with participant-cluster-robust 95% confidence intervals, FDR-adjusted p values, and exact
1272 support. The analysis includes 17,935 participant-hours from 140 participants, covering 801
1273 participant-days and nine sites. Panels A–C and panel D answer different questions and are
1274 not numerically interchangeable. Estimates are observational. During reported sleep,
1275 measurements describe the bedside sleep environment.

1276 Hourly routine analyses

1277 **Supplementary Figure S9. Paired sensor-position hourly** 1278 **associations**

1279

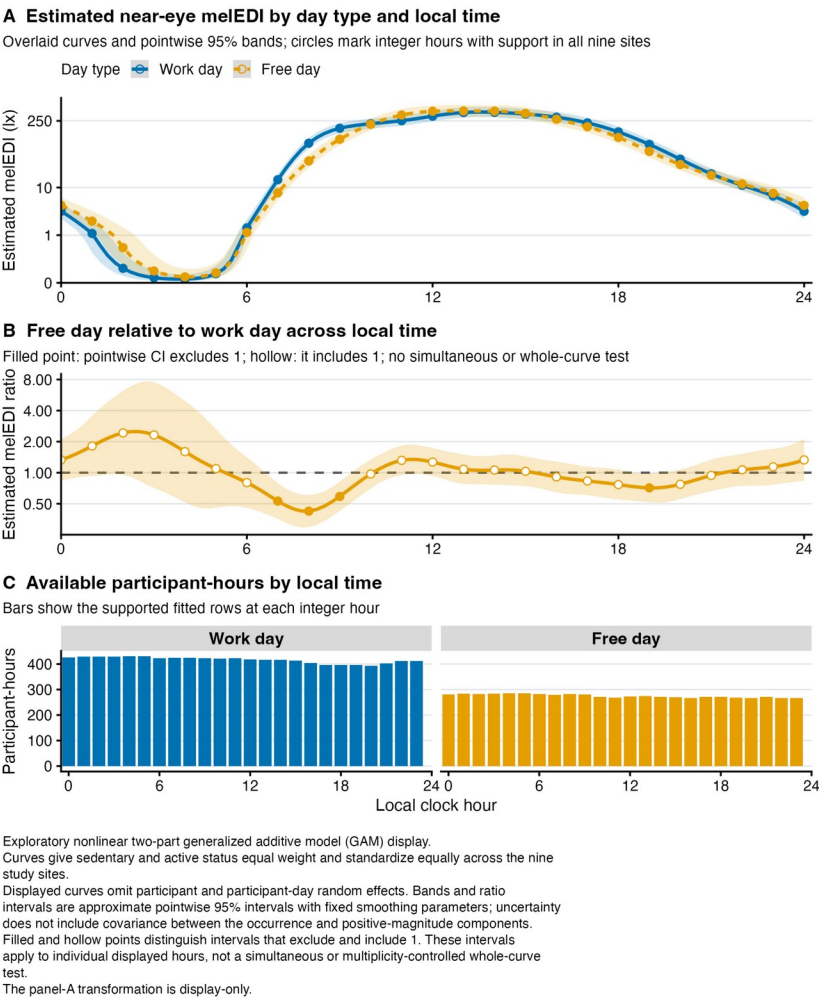


1280

1281 Supplementary Figure S9. Near-eye and complementary chest associations estimated
1282 separately for the same participants and participant-days. The sets of supported participant-
1283 hours differ between sensor positions, so the display is not an observation-level identity
1284 comparison and does not test equivalence.

1285 **Supplementary Figure S10. Free-versus-Work hourly timing**

1286

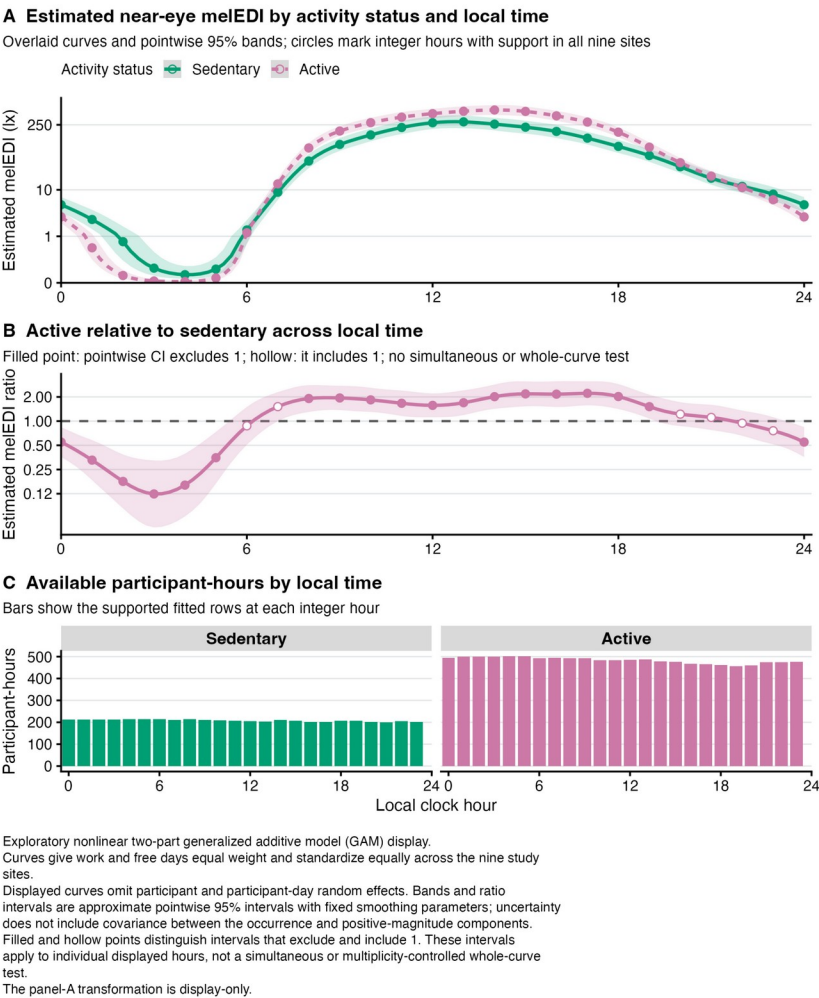


1287

1288 Supplementary Figure S10. Exploratory Free-versus-Work hourly fitted exposure and
1289 response-scale ratio curves with 95% confidence intervals and participant-hour support.

1290 **Supplementary Figure S11. Active-versus-Sedentary hourly timing**

1291

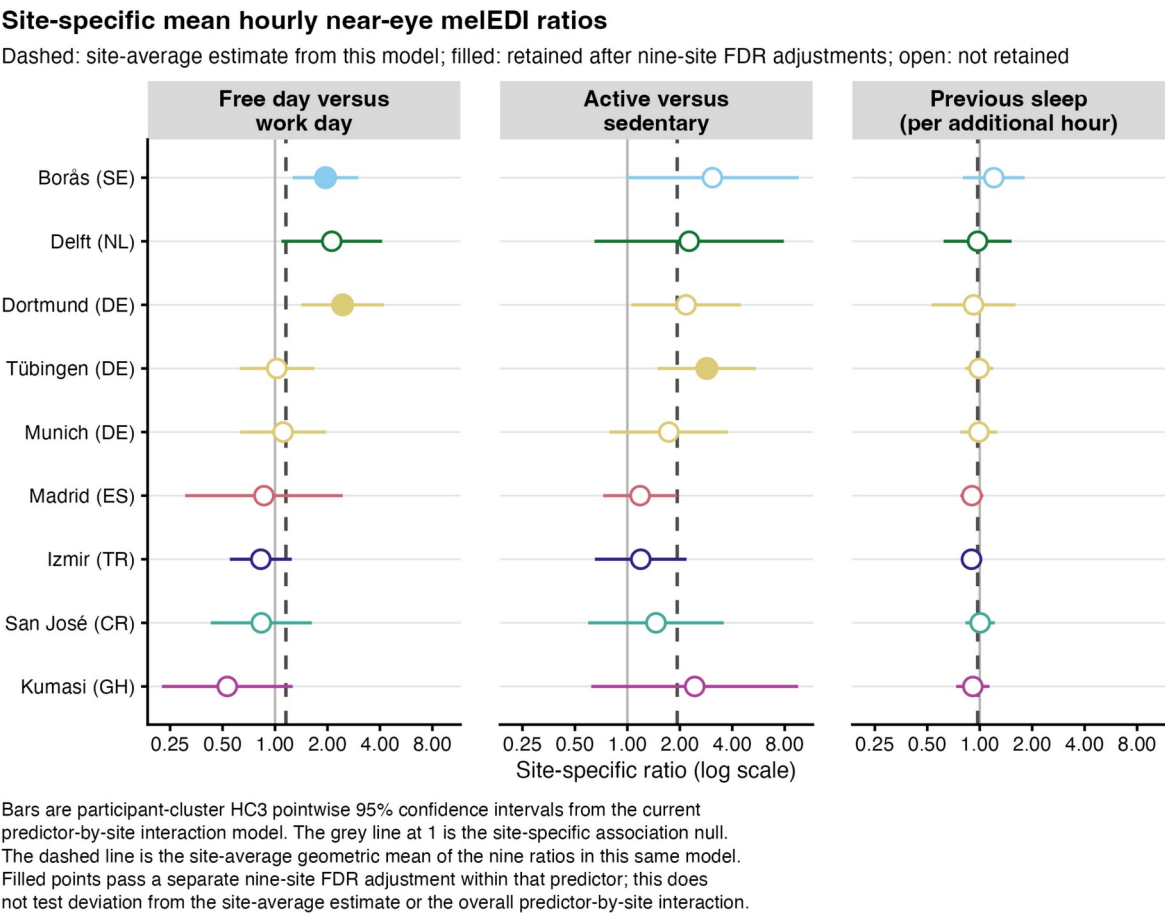


1292

1293 Supplementary Figure S11. Exploratory Active-versus-Sedentary hourly fitted exposure and
1294 response-scale ratio curves with 95% confidence intervals and participant-hour support.

1295 **Supplementary Figure S12. Site-specific hourly routine**
1296 **associations**

1297



1298

1299 Supplementary Figure S12. Site-specific estimates from the predictor-by-site interaction

1300 model. Filled and hollow points distinguish associations that did and did not meet the 0.050

1301 criterion after a separate nine-site FDR adjustment within each predictor. The 95%

1302 confidence intervals are unadjusted; dashed lines show site-average estimates calculated

1303 with equal weight for each of the nine sites. Sites are components of one pooled model, not

1304 independent replications or causal effects of location.

1305 **Supplementary Table S9. Participant-hour routine associations**

Predictor	Ratio (95% CI)	FDR-adjusted p
Free day versus work day	1.45 × (1.14–1.85)	0.004
Active versus sedentary	2.06 × (1.54–2.76)	<0.001
Previous sleep duration (per hour)	0.98 × (0.85–1.12)	0.745
Directions are free versus work day, active versus sedentary day status, and the ratio per one additional hour of previous sleep. Participant-cluster-robust 95% CIs use HC3 with 137 participants. The three p-values form one FDR-adjusted family; the null ratio is 1.		

1306

1307 **Supplementary Table S9.** Common-association estimates from the participant-hour analysis of Free versus Work day, Active versus Sedentary daily activity
1308 status and each additional hour of previous-night sleep duration. Ratios and participant-cluster-robust 95% confidence intervals come from the fixed-site model;
1309 the three *p* values form one FDR-adjusted family. Predictor-by-site differences are shown in Supplementary Figure S12.

1310 **Person-level findings**

1311 **Supplementary Table S10. Person-level evidence synthesis**

Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Light-exposure behaviour and awareness factors	68 factor-by-metric cells across four participant-level factors	No cell retained FDR support; four sleep-environment cells were non-estimable	68 near-eye factor-by-metric tests	None retained	Metric-specific near-eye samples in Supplementary Table S11	Broad inconclusive result; non-retention is not evidence of no association
Visual light sensitivity	Nine near-eye metrics per one VLSQ-8 SD	Largest displayed ratio shift was melanopic EDI dose, $\times 0.846$ (0.716 to 0.999)	Nine near-eye metric tests	None retained; adjusted $p = 0.160$ for this row	141 participants, 761 participant-days, nine sites for the displayed row	Broad inconclusive result
Corrected midsleep on Free days	Timing outcomes per one-hour later MSFsc	M10 midpoint +0.205 h (0.045 to 0.366); L10 midpoint +0.276 h (0.118 to 0.435); first light above 250 lx +0.381 h (0.146 to 0.616)	Five MCTQ timing outcomes, separate from MEQ	These three rows retained	139 to 140 participants; 722 to 810 participant-days; nine sites	Selective timing associations
Morningness-Eveningness Questionnaire	Timing outcomes per 10 points greater morning preference	M10 midpoint -0.278 h (-0.451 to -0.105); L10 midpoint -0.314 h (-0.487 to -0.142); first light above 250 lx -0.444 h (-0.699 to -0.189)	Five MEQ timing outcomes, separate from MCTQ	These three rows retained	140 to 141 participants; 727 to 816 participant-days; nine sites	Selective timing associations

Table continued

Person-level evidence synthesis						
	Outcome or analysis scale	Result	FDR-correction set	FDR decision	Exact sample	Interpretation and qualification
Age	Near-eye daily metrics per 10 years	Brightest 10 h geometric mean $\times 1.31$ (1.09 to 1.58); time above 1,000 lx $\times 1.27$ (1.11 to 1.45); melanopic EDI dose $\times 1.31$ (1.09 to 1.56)	Near-eye age-metric tests	Three rows retained	141 participants; 761 to 816 participant-days; nine sites	Selective age associations; complementary chest results remain separate
Metric-level biological sex	Near-eye daily metrics, Female minus Male	No near-eye metric-level result retained FDR support	Near-eye biological-sex metric tests	None retained	Metric-specific samples in Supplementary Table S14	Biological sex and gender were recorded separately; gender was not analysed
Biological-sex-specific daily curve	Complete 24-hour curve and activity-complete sensitivity	The all-available near-eye complete-curve test was supported, but separation attenuated and was not retained in the activity-complete near-eye analysis	Global complete-curve tests and separately prespecified sensitivity decisions	Analysis-specific global decisions	All-available near eye: 141 participants, 816 participant-days, 37,756 observations, nine sites	Attenuation-first interpretation; confidence intervals apply at the displayed clock times

Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. The two chronotype instruments were analysed separately. Biological sex and gender were recorded separately; gender was not analysed.

1313

1314 **Supplementary Table S10.** Summary of person-level results in the near-eye analyses. Each row identifies the analysis-specific FDR-correction set or decision

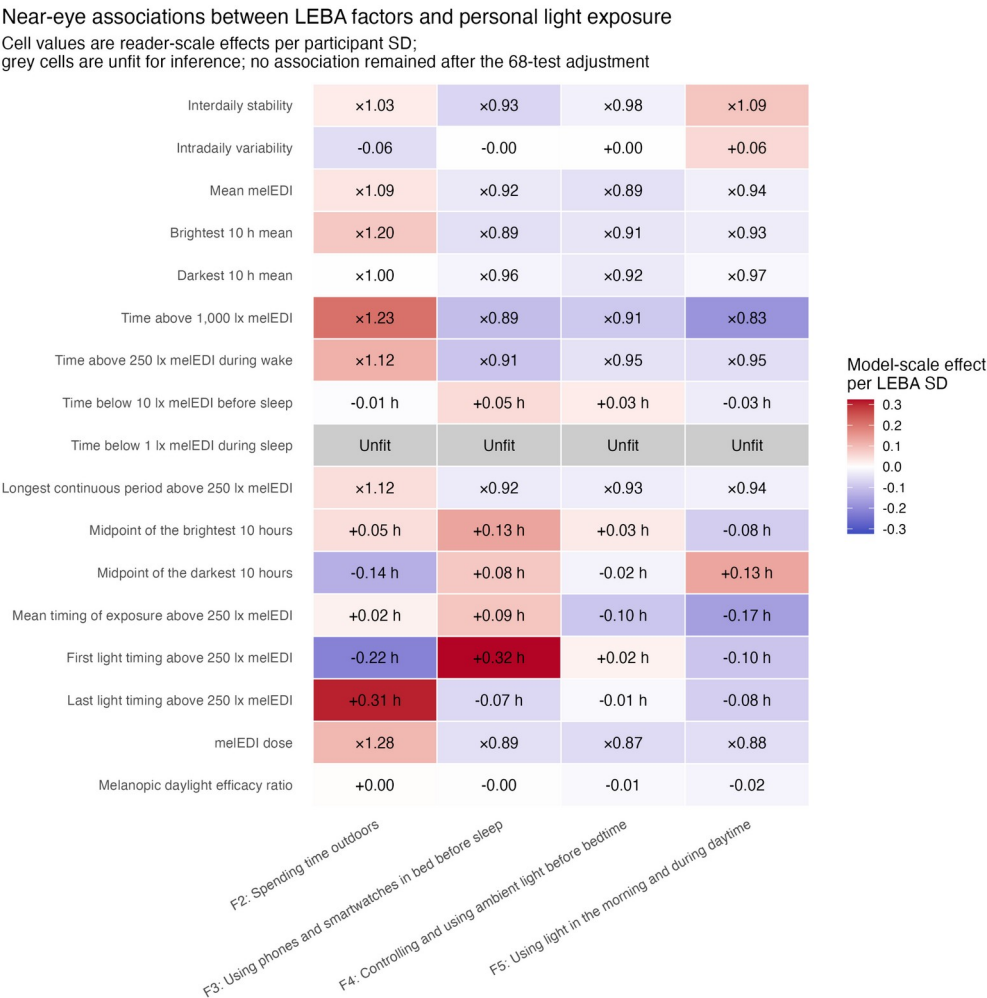
1315 structure applied to that result. Results that did not meet the applicable FDR criterion are inconclusive rather than evidence of no association. Four light-

1316 behaviour sleep-environment cells were non-estimable. The two chronotype instruments were analysed separately. Biological sex and gender were recorded

1317 separately; gender was not analysed. The complete available-data curve difference was attenuated and not retained in the activity-complete near-eye analysis.

1318 **Supplementary Figure S13 and Table S11. Light-exposure**
1319 **behaviour and awareness**

1320



1321

1322 Supplementary Figure S13. Near-eye associations per one participant-level standard
1323 deviation of each Light Exposure Behaviour Assessment factor. None of the 68 tests
1324 retained FDR-adjusted support. Four sleep-environment cells were not estimable and are
1325 labelled Unfit with estimates suppressed. Lack of adjusted support is inconclusive rather
1326 than proof of no association.

	F2	F3	F4	F5
Interdaily stability	OR 1.028 (0.956–1.106) FDR-adjusted p = 0.658	OR 0.931 (0.868–0.999) FDR-adjusted p = 0.260	OR 0.975 (0.906–1.050) FDR-adjusted p = 0.679	OR 1.089 (1.011–1.174) FDR-adjusted p = 0.202
Intradaily variability	-0.057 (-0.124–+0.009) FDR-adjusted p = 0.351	-0.001 (-0.068–+0.065) FDR-adjusted p = 0.978	+0.001 (-0.068–+0.069) FDR-adjusted p = 0.978	+0.057 (-0.013–+0.127) FDR-adjusted p = 0.362
Mean melEDI	×1.092 (0.965–1.235) FDR-adjusted p = 0.379	×0.917 (0.813–1.034) FDR-adjusted p = 0.379	×0.891 (0.787–1.009) FDR-adjusted p = 0.313	×0.941 (0.827–1.070) FDR-adjusted p = 0.585
Brightest 10 h mean	×1.204 (1.008–1.439) FDR-adjusted p = 0.260	×0.888 (0.745–1.058) FDR-adjusted p = 0.384	×0.906 (0.756–1.087) FDR-adjusted p = 0.545	×0.931 (0.772–1.123) FDR-adjusted p = 0.658
Darkest 10 h mean	×0.995 (0.905–1.094) FDR-adjusted p = 0.958	×0.956 (0.871–1.049) FDR-adjusted p = 0.577	×0.921 (0.838–1.013) FDR-adjusted p = 0.351	×0.970 (0.879–1.070) FDR-adjusted p = 0.716
Time above 1,000 lx melEDI	×1.234 (1.084–1.405) FDR-adjusted p = 0.124	×0.895 (0.785–1.020) FDR-adjusted p = 0.362	×0.911 (0.797–1.043) FDR-adjusted p = 0.390	×0.831 (0.725–0.953) FDR-adjusted p = 0.169
Time above 250 lx melEDI during wake	×1.122 (1.006–1.252) FDR-adjusted p = 0.260	×0.907 (0.814–1.010) FDR-adjusted p = 0.351	×0.953 (0.852–1.065) FDR-adjusted p = 0.614	×0.946 (0.843–1.061) FDR-adjusted p = 0.586
Time below 10 lx melEDI before sleep	-0.005 h (-0.135–+0.124 h) FDR-adjusted p = 0.958	+0.052 h (-0.073–+0.177 h) FDR-adjusted p = 0.614	+0.028 h (-0.104–+0.159 h) FDR-adjusted p = 0.781	-0.031 h (-0.167–+0.105 h) FDR-adjusted p = 0.781
Time below 1 lx melEDI during sleep	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display	Unfit for inference Estimate and p-values suppressed from the reader display
Longest continuous period above 250 lx melEDI	×1.117 (1.007–1.238) FDR-adjusted p = 0.260	×0.924 (0.835–1.023) FDR-adjusted p = 0.375	×0.933 (0.840–1.037) FDR-adjusted p = 0.393	×0.945 (0.847–1.053) FDR-adjusted p = 0.557

Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.

1329

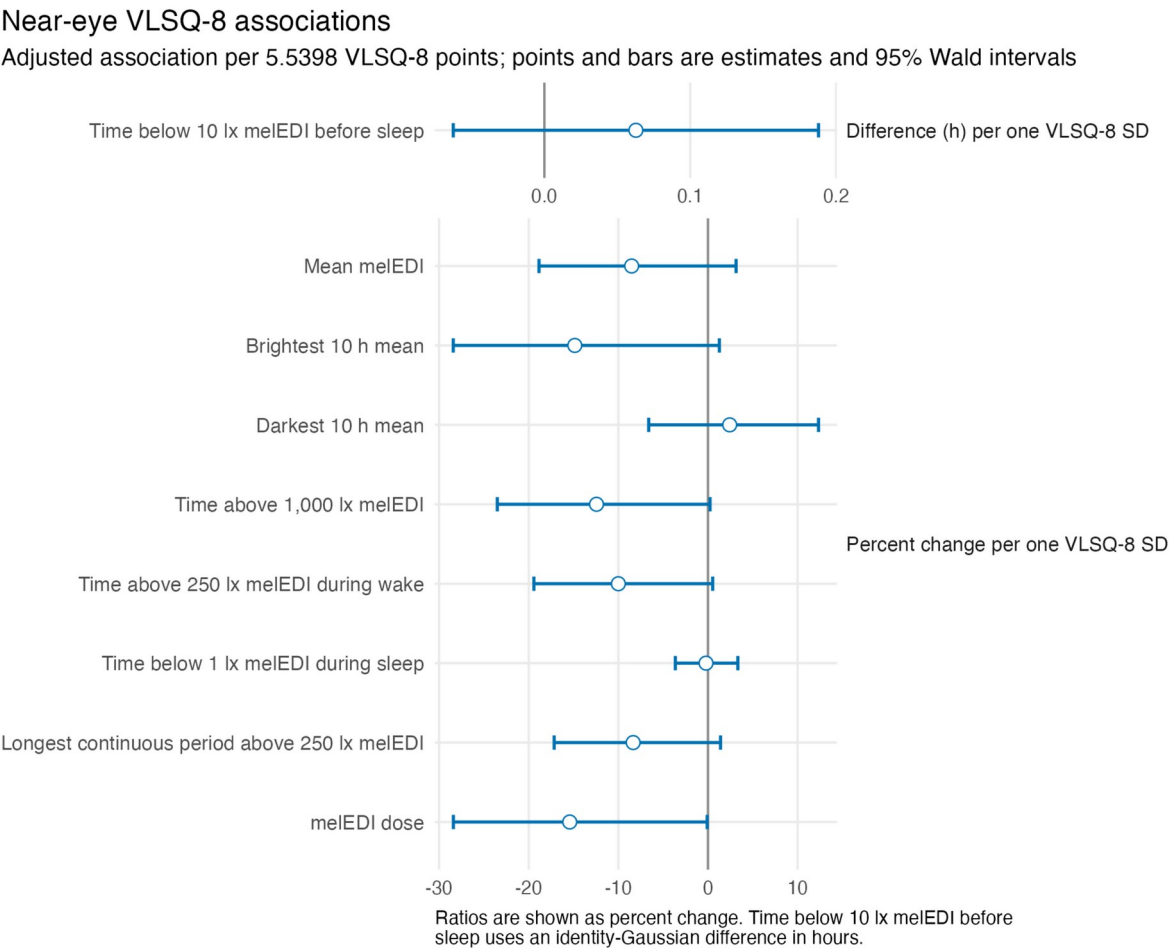
1330 *Supplementary Table S11 (continued).*

	F2	F3	F4	F5
Midpoint of the brightest 10 hours	+0.049 h (-0.132–+0.229 h) FDR-adjusted p = 0.747	+0.133 h (-0.042–+0.309 h) FDR-adjusted p = 0.375	+0.030 h (-0.153–+0.213 h) FDR-adjusted p = 0.868	-0.083 h (-0.270–+0.105 h) FDR-adjusted p = 0.601
Midpoint of the darkest 10 hours	-0.136 h (-0.317–+0.044 h) FDR-adjusted p = 0.375	+0.080 h (-0.097–+0.258 h) FDR-adjusted p = 0.594	-0.020 h (-0.204–+0.164 h) FDR-adjusted p = 0.935	+0.132 h (-0.056–+0.320 h) FDR-adjusted p = 0.379
Mean timing of exposure above 250 lx melEDI	+0.020 h (-0.152–+0.192 h) FDR-adjusted p = 0.935	+0.086 h (-0.081–+0.253 h) FDR-adjusted p = 0.558	-0.097 h (-0.267–+0.074 h) FDR-adjusted p = 0.510	-0.166 h (-0.339–+0.008 h) FDR-adjusted p = 0.301
First light timing above 250 lx melEDI	-0.219 h (-0.489–+0.052 h) FDR-adjusted p = 0.362	+0.324 h (+0.062–+0.585 h) FDR-adjusted p = 0.179	+0.019 h (-0.252–+0.291 h) FDR-adjusted p = 0.958	-0.097 h (-0.380–+0.185 h) FDR-adjusted p = 0.679
Last light timing above 250 lx melEDI	+0.308 h (+0.043–+0.573 h) FDR-adjusted p = 0.202	-0.066 h (-0.328–+0.196 h) FDR-adjusted p = 0.760	-0.011 h (-0.280–+0.259 h) FDR-adjusted p = 0.958	-0.081 h (-0.361–+0.200 h) FDR-adjusted p = 0.743
melEDI dose	×1.278 (1.079–1.514) FDR-adjusted p = 0.124	×0.885 (0.748–1.048) FDR-adjusted p = 0.379	×0.872 (0.732–1.038) FDR-adjusted p = 0.375	×0.881 (0.736–1.054) FDR-adjusted p = 0.379
Melanopic daylight efficacy ratio	+0.004 (-0.011–+0.019) FDR-adjusted p = 0.747	-0.004 (-0.019–+0.011) FDR-adjusted p = 0.747	-0.010 (-0.026–+0.005) FDR-adjusted p = 0.380	-0.020 (-0.035–+0.004) FDR-adjusted p = 0.169
Associations are reported per one participant-level SD of the LEBA score with 95% CIs. FDR-adjusted p is bold only at alpha = 0.050 across all 68 tests. Sleep-environment cells are unfit for inference and suppressed.				

1331

1332 **Supplementary Figure S14 and Table S12. Visual light sensitivity**

1333



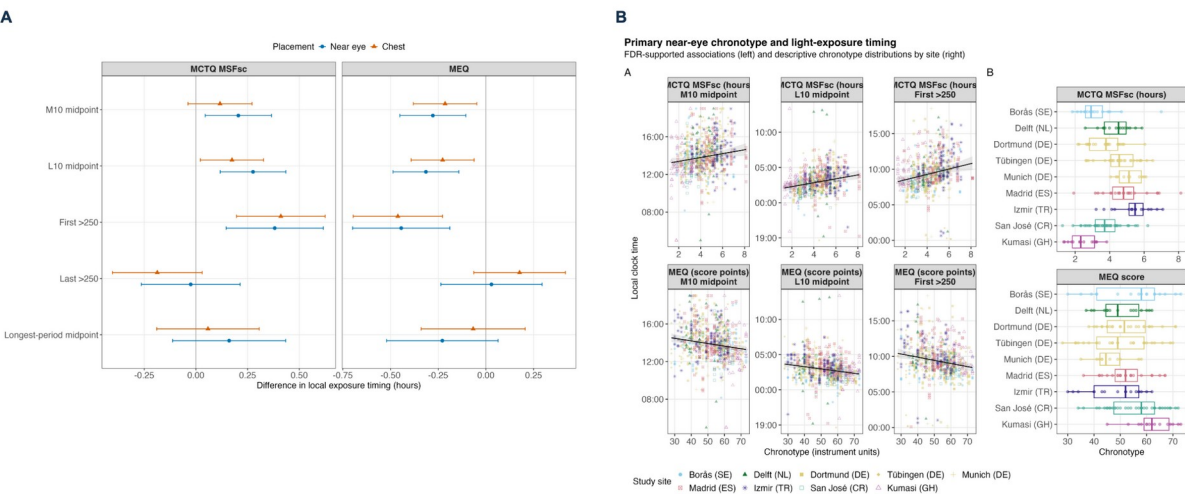
1334

1335 Supplementary Figure S14. Near-eye site-average associations between the Visual Light
1336 Sensitivity Questionnaire-8 and nine personal light-exposure metrics. Associations compare
1337 scores separated by one participant-level standard deviation. Horizontal bars are 95% Wald
1338 confidence intervals. None of the nine associations retained FDR-adjusted support.

Metric	Scale	Association per VLSQ-8 SD (95% CI)	FDR-adjusted p	Participants / participant-days / sites
Mean melEDI	Ratio (percentage change)	×0.915 (0.811–1.031)	0.221	141 / 816 / 9
Brightest 10 h mean	Ratio (percentage change)	×0.851 (0.716–1.013)	0.160	141 / 816 / 9
Darkest 10 h mean	Ratio (percentage change)	×1.024 (0.934–1.123)	0.689	141 / 816 / 9
Time above 1,000 lx melEDI	Ratio (percentage change)	×0.876 (0.765–1.002)	0.160	141 / 816 / 9
Time above 250 lx melEDI during wake	Ratio (percentage change)	×0.900 (0.806–1.005)	0.160	141 / 737 / 9
Time below 10 lx melEDI before sleep	Difference in hours	+0.063 h (-0.062–+0.188 h)	0.422	139 / 655 / 9
Time below 1 lx melEDI during sleep	Ratio (percentage change)	×0.998 (0.964–1.033)	0.903	141 / 778 / 9
Longest continuous period above 250 lx melEDI	Ratio (percentage change)	×0.917 (0.828–1.014)	0.167	141 / 816 / 9
melEDI dose	Ratio (percentage change)	×0.846 (0.716–0.999)	0.160	141 / 761 / 9
Associations compare scores separated by one participant-level SD (5.540 VLSQ-8 points). The hour-scale row is an absolute difference; ratio rows can be read as percentage changes. Intervals are two-sided 95% Wald CIs. FDR adjustment is across all nine metrics; adjusted p-values would be bold at 0.050.				

1341 **Supplementary Figure S15 and Table S13. Chronotype and timing**

1342



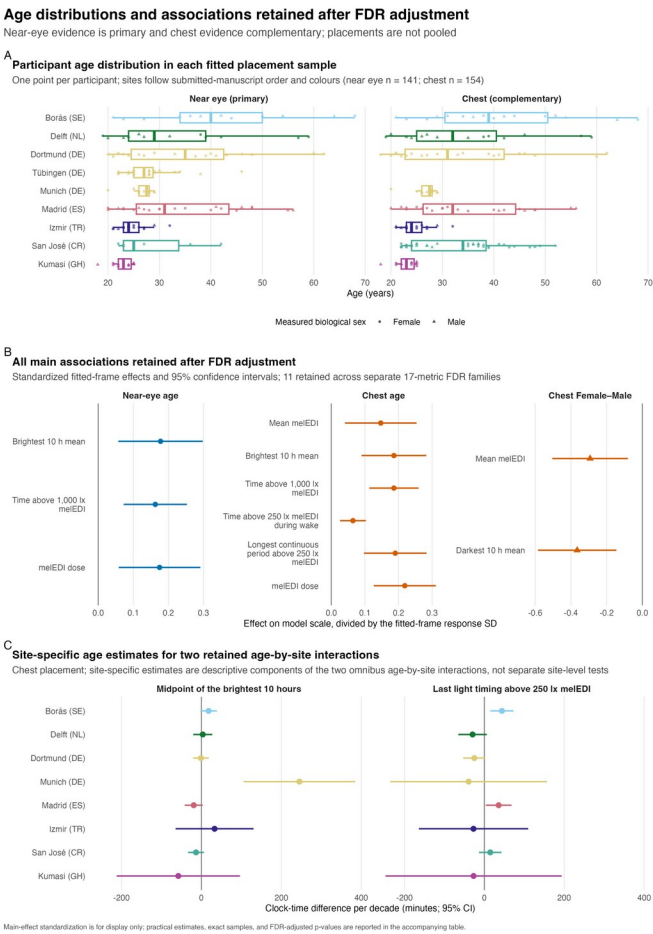
1343

1344 Supplementary Figure S15. Chronotype and timing of personal light exposure. A, Study-site-
1345 adjusted associations of sleep-timing-based corrected midsleep on Free days and
1346 questionnaire-based morningness-eveningness with the analysed timing metrics. Points are
1347 estimates and bars are 95% confidence intervals; the two chronotype instruments remain
1348 separate. B, Observed participant-day timing values across chronotype and country-coded
1349 study sites for the samples used in the corresponding models. Each point represents one
1350 participant-day in a fitted sample. The observed panel is descriptive and does not replace
1351 the adjusted models or estimate independent site effects. Near-eye estimates provide
1352 ocular-exposure evidence; chest estimates, where shown, are complementary non-ocular
1353 evidence.

Metric	Instrument	Signed association, h (95% CI)	FDR-adjusted p	Fitted sample: participants / days / observations / hours / sites
Midpoint of the brightest 10 hours	MCTQ MSFsc	+0.205 (+0.045 to +0.366)	0.017	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the brightest 10 hours	MEQ	-0.278 (-0.451 to -0.105)	0.002	141 / 816 / 816 / 18851.0 h / 9
Midpoint of the darkest 10 hours	MCTQ MSFsc	+0.276 (+0.118 to +0.435)	0.003	140 / 810 / 810 / 18711.7 h / 9
Midpoint of the darkest 10 hours	MEQ	-0.314 (-0.487 to -0.142)	0.001	141 / 816 / 816 / 18851.0 h / 9
First light timing above 250 lx meLEDI	MCTQ MSFsc	+0.381 (+0.146 to +0.616)	0.003	139 / 722 / 722 / 16716.3 h / 9
First light timing above 250 lx meLEDI	MEQ	-0.444 (-0.699 to -0.189)	0.001	140 / 727 / 727 / 16832.5 h / 9
Last light timing above 250 lx meLEDI	MCTQ MSFsc	-0.025 (-0.264 to +0.214)	0.811	140 / 683 / 683 / 15900.8 h / 9
Last light timing above 250 lx meLEDI	MEQ	+0.029 (-0.236 to +0.294)	0.814	141 / 687 / 687 / 15995.0 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MCTQ MSFsc	+0.161 (-0.113 to +0.434)	0.284	131 / 478 / 478 / 11325.5 h / 9
Midpoint of the longest continuous period above 250 lx meLEDI	MEQ	-0.228 (-0.521 to +0.064)	0.135	132 / 482 / 482 / 11419.8 h / 9
MCTQ associations are per one-hour later MSFsc; MEQ associations are per 10 points greater morning preference. Positive associations indicate later timing and negative associations earlier. Intervals are 95% Wald CIs. FDR-adjusted p-values are bold when FDR-adjusted p < 0.050 within the instrument-specific five-outcome family.				

1356 **Supplementary Figure S16 and Table S14. Age and biological sex**

1357



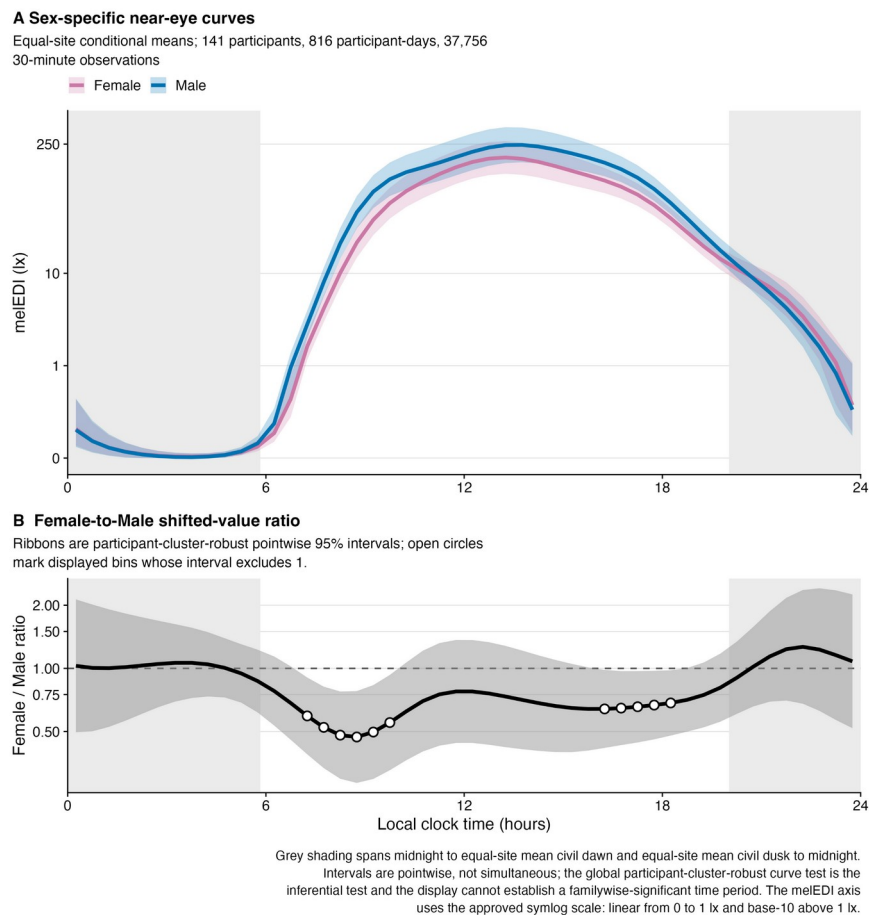
1358

1359 Supplementary Figure S16. Participant age distributions, the 11 main associations retained
1360 after FDR adjustment, and descriptive site-specific estimates for two retained age-by-site
1361 interactions in the complementary chest analysis. The site-specific interaction estimates are
1362 descriptive model contrasts, not independent site tests. Biological sex, coded Female or
1363 Male, was analysed; gender was recorded separately and was not analysed.

Placement and role	Predictor	Metric	Practical association (95% CI)	FDR-adjusted p	Sample
Near eye (primary)	Age, per 10 years	Brightest 10 h mean	1.31× (1.09–1.58)	0.018	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	Time above 1,000 lx melEDI	1.27× (1.11–1.45)	0.009	141 participants; 816 participant-days/observations
Near eye (primary)	Age, per 10 years	melEDI dose	1.31× (1.09–1.56)	0.018	141 participants; 761 participant-days/observations
Chest (complementary)	Age, per 10 years	Mean melEDI	1.16× (1.04–1.30)	0.017	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Brightest 10 h mean	1.35× (1.15–1.57)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 1,000 lx melEDI	1.31× (1.18–1.46)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	Time above 250 lx melEDI during wake	1.16× (1.06–1.26)	0.004	154 participants; 818 participant-days/observations
Chest (complementary)	Age, per 10 years	Longest continuous period above 250 lx melEDI	1.18× (1.09–1.27)	<0.001	154 participants; 902 participant-days/observations
Chest (complementary)	Age, per 10 years	melEDI dose	1.40× (1.21–1.61)	<0.001	154 participants; 851 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Mean melEDI	0.74× (0.60–0.92)	0.048	154 participants; 902 participant-days/observations
Chest (complementary)	Measured biological sex, Female minus Male	Darkest 10 h mean	0.76× (0.65–0.90)	0.016	154 participants; 902 participant-days/observations
Associations are site-adjusted estimates per 10-year age increase or Female minus Male. Ratios are back-transformed to the practical scale. Each FDR-adjusted p-value belongs to its sensor-position- and predictor-specific complete 17-test family; bold values meet the FDR-adjusted p < 0.050 rule. Samples give the exact fitted participants and participant-day observations for these retained daily-response results.					

1366 **Supplementary Figure S17 and Table S15. Biological-sex-specific**
1367 **daily curves**

1368



1369

1370 Supplementary Figure S17. Near-eye biological-sex-specific fitted melanopic equivalent
1371 daylight illuminance curves and Female-to-Male shifted-value ratio. Ribbons and contrast
1372 intervals are participant-cluster-robust 95% confidence intervals at the displayed clock times.
1373 Grey shading gives site-average civil-night context with each site weighted equally and is not
1374 a model covariate. Biological sex, coded Female or Male, was analysed; gender was
1375 recorded separately and was not analysed.

	Role	Test	F statistic	Numerator df	Denominator df	Raw p	FDR-adjusted p	Support
Near eye	Primary	Complete Female-minus-Male 24-hour curve	20.946	9.692	131	0.028	0.028	Supported
Chest	Complementary	Complete Female-minus-Male 24-hour curve	4.876	1.003	152	0.029	0.029	Supported

Primary near-eye fitted sample: 141 participants, 816 participant-days, 37,756 30-minute observations, and 9 sites. Complementary chest fitted sample: 154 participants, 902 participant-days, 41,842 30-minute observations, and 8 sites.

Each row is one joint complete-curve test. Raw p is bold at raw $p < 0.050$; FDR-adjusted p is bold independently at adjusted $p < 0.050$. Each labelled global family contains one test, so the stored raw and adjusted values coincide.

1378 **Measurement position, robustness, and reproducibility**

1379 Near-eye and complementary chest measurements remain separate sensor-position
1380 estimands. Common-sample analyses use the same participants and participant-days at
1381 both positions and fit the positions separately. They do not pool positions, test equivalence,
1382 or establish a universal correction. Detailed model checks, sensitivities, preregistration
1383 deviations, exact formulas, package versions, and source documentation are provided in the
1384 analysis reports and reproducibility package.